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ENVIRONMENT

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1 Problem 1: Introduction to OPF

1.1 Task 1-1: Method Approximation

DC Optimal Power Flow

What it does

DC Optimal Power Flow (DC-OPF) computes how power should be produced and flow through the network in the cheapest possible way, while respecting physical network constraints. The model uses simplified versions of Kirchhoff's laws, meaning that when a generator increases its output, the resulting flow distributes automatically across all lines based on their impedance, just as it does in reality.

Key considerations

Line capacities, generator limits, and the fact that line flows are determined endogenously by the model itself must all be accounted for. Network topology and impedance values must therefore be known inputs.

Transport Network (ATC)

What it does

The Available Transfer Capability (ATC) approach treats the network as simple pipes between zones. A fixed limit is set for how much power can flow from zone A to zone B, and the market optimises trade within these limits. The model does not account for how flows physically distribute themselves within the network.

Key considerations

ATC values are set exogenously by a network operator in advance, and must be conservative enough to account for uncertainty related to loop flows and other physical effects that the model does not capture.

The Key Difference

The fundamental difference lies in how the network constraints are handled:

- **DC-OPF** uses the physics of the network. The power flow is governed by $P_{ij} = b_{ij}(\delta_i - \delta_j)$, meaning the model accounts for parallel path flows (loop flows).
- **ATC** treats the grid as a commercial transport network where P_{ij} is only constrained by

a pre-defined thermal limit $P_{ij,max}$.

This leads to "Parallel Path Flow" issues in ATC, where a trade between two nodes physically affects lines in the rest of the system, which is something the ATC model ignores, but the DC-OPF captures accurately.

1.2 Task 1-2: Lagrangian Expression of DCOPF

The objective is to minimise total generation cost subject to the power-balance constraint. With no line or capacity constraints the Lagrangian is

$$\mathcal{L}(P_{G1}, P_{G2}, P_{G3}, \lambda) = \sum_{n=1}^3 c_n P_{Gn} + \lambda \left(P_{\text{load,tot}} - \sum_{n=1}^3 P_{Gn} \right)$$

The total load is

$$P_{\text{load,tot}} = 200 + 200 + 500 = 900 \text{ MW}$$

Substituting the generation costs ($c_1 = 300$, $c_2 = 1000$, $c_3 = 600$ NOK/MWh) and the power-balance equality constraint gives

$$\mathcal{L} = 300 P_{G1} + 1000 P_{G2} + 600 P_{G3} + \lambda (900 - P_{G1} - P_{G2} - P_{G3})$$

where λ is the Lagrange multiplier representing the system marginal price (NOK/MWh).

1.3 Task 1-3: DCOPF Without Line Constraints

Optimal generation

Dispatch follows the *merit order* (cheapest generator first):

1. Generator 1 is cheapest at 300 NOK/MWh (capacity 1000 MW). It covers the entire load of 900 MW on its own.

$$P_{G1} = 900 \text{ MW}, \quad P_{G2} = 0 \text{ MW}, \quad P_{G3} = 0 \text{ MW}$$

System (marginal) price

The marginal generator is Generator 1, so

$$\lambda = c_1 = 300 \text{ NOK/MWh}$$

Total system cost

$$TC = 900 \times 300 = 270\,000 \text{ NOK}$$

Phase angles

Using $S_{\text{base}} = 100 \text{ MVA}$, the net nodal injections in per unit are

$$P_1 = \frac{900 - 200}{100} = 7 \text{ p.u.}, \quad P_2 = \frac{0 - 200}{100} = -2 \text{ p.u.}, \quad P_3 = \frac{0 - 500}{100} = -5 \text{ p.u.}$$

With $\delta_1 = 0$ as the reference, the nodal power balance $P_n = \sum_l b_{nl}(\delta_n - \delta_l)$ gives two equations:

Node 2 ($b_{12} = 20$, $b_{23} = 30$):

$$-2 = b_{12}(\delta_2 - 0) + b_{23}(\delta_2 - \delta_3) = 20\delta_2 + 30(\delta_2 - \delta_3) \Rightarrow 50\delta_2 - 30\delta_3 = -2$$

Node 3 ($b_{13} = 10$, $b_{23} = 30$):

$$-5 = b_{13}(\delta_3 - 0) + b_{32}(\delta_3 - \delta_2) = 10\delta_3 + 30(\delta_3 - \delta_2) \Rightarrow -30\delta_2 + 40\delta_3 = -5$$

From the first equation:

$$\delta_2 = \frac{-2 + 30\delta_3}{50} = -0.04 + 0.6\delta_3$$

Substituting into the second equation:

$$-30(-0.04 + 0.6\delta_3) + 40\delta_3 = -5 \Rightarrow 1.2 - 18\delta_3 + 40\delta_3 = -5 \Rightarrow 22\delta_3 = -6.2$$

$$\delta_3 = -\frac{6.2}{22} = -\frac{31}{110} \approx -0.282 \text{ rad}$$

$$\delta_2 = -0.04 + 0.6 \times (-0.282) = -0.04 - 0.169 \approx -0.209 \text{ rad}$$

$$\delta_1 = 0, \quad \delta_2 \approx -0.209 \text{ rad}, \quad \delta_3 \approx -0.282 \text{ rad}$$

Power flows (in per unit, then convert)

Power flows are defined as $P_{ij} = b_{ij}(\delta_i - \delta_j)$, where a positive value means flow from bus i to bus j .

$$P_{12} = b_{12}(\delta_1 - \delta_2) = 20(0 - (-0.209)) = 20 \times 0.209 \approx 4.18 \text{ p.u.} \Rightarrow 418 \text{ MW (bus 1} \rightarrow \text{bus 2)}$$

$$P_{13} = b_{13}(\delta_1 - \delta_3) = 10(0 - (-0.282)) = 10 \times 0.282 \approx 2.82 \text{ p.u.} \Rightarrow 282 \text{ MW (bus 1} \rightarrow \text{bus 3)}$$

$$P_{23} = b_{23}(\delta_2 - \delta_3) = 30(-0.209 - (-0.282)) = 30 \times 0.073 \approx 2.18 \text{ p.u.} \Rightarrow 218 \text{ MW (bus 2} \rightarrow \text{bus 3)}$$

Nodal balance check:

$$\text{Bus 1: } P_{12} + P_{13} = 4.18 + 2.82 = 7.00 \text{ p.u.} = P_1 \checkmark$$

$$\text{Bus 2: } -P_{12} + P_{23} = -4.18 + 2.18 = -2.00 \text{ p.u.} = P_2 \checkmark$$

$$\text{Bus 3: } -P_{13} - P_{23} = -2.82 - 2.18 = -5.00 \text{ p.u.} = P_3 \checkmark$$

This is physically consistent: Generator 1 at bus 1 produces all the power, which flows outward to the load buses. Bus 2 receives 418 MW, consumes 200 MW locally, and forwards the remaining 218 MW onward to bus 3.

Nodal prices

Since there are no binding transmission constraints, all nodes see the same marginal price:

$$\lambda_1 = \lambda_2 = \lambda_3 = 300 \text{ NOK/MWh}$$

1.4 Task 1-4: DCOPF With Line Constraints

The new thermal limits are $P_{12}^{\max} = 500 \text{ MW}$, $P_{13}^{\max} = 500 \text{ MW}$, $P_{23}^{\max} = 100 \text{ MW}$.

(a) Qualitative effect of line constraints

Without line constraints, Generator 1 (cheapest at 300 NOK/MWh) would supply the entire 900 MW load, leading to $P_{23} \approx 218$ MW (see Task 1-3 flows). This violates the 100 MW limit on Line 2-3.

The binding constraint on Line 2-3 prevents the unconstrained-optimal power pattern from being realised. As a result, Generator 3 at Node 3 must pick up part of the local load rather than relying on cheap remote power, and the system marginal price is no longer uniform: nodes separated by the congested line will see different nodal prices. Lines 1-2 and 1-3 are not binding at the constrained optimum.

(b) Additional term in the Lagrangian

For each transmission line ij with an active upper flow limit, a non-negative multiplier $\mu_{ij} \geq 0$ is added. With only Line 2-3 binding:

$$\mathcal{L} = \sum_{n=1}^3 c_n P_{Gn} + \lambda \left(\sum_n D_n - \sum_n P_{Gn} \right) + \mu_{23} (P_{23} - P_{23}^{\max})$$

where λ is the system energy price (dual of the power balance) and $\mu_{23} \geq 0$ is the shadow price (congestion rent) of Line 2-3. The DC flow is $P_{23} = b_{23}(\delta_2 - \delta_3)$, which can be expressed directly in terms of generator outputs via the B-matrix (see below).

(c) First-order conditions

Taking partial derivatives and setting them to zero:

$$\frac{\partial \mathcal{L}}{\partial P_{Gi}} = 0 \Rightarrow \lambda_i = c_i + (\text{line constraint contributions})$$

For a node that is not separated by a congested line the nodal price equals the system marginal price. When a line is congested, prices on either side differ:

$$\lambda_i \neq \lambda_j \quad \text{when line } ij \text{ is binding}$$

The shadow price γ_{23} captures this price difference:

$$\gamma_{23} = \lambda_2 - \lambda_3$$

(d) Numerical solution

Optimal generation. Without any line limit, $P_{G2} = P_{G3} = 0$ and $P_{23} \approx 218$ MW, which violates $P_{23}^{\max} = 100$ MW. The constraint on Line 2-3 is therefore binding at the optimum.

Since Generator 2 is the most expensive unit (1000 NOK/MWh), the cost is minimised by setting $P_{G2} = 0$ and distributing the remaining generation between G_1 (300 NOK/MWh) and G_3 (600 NOK/MWh).

With $P_{G2} = 0$, the DC flow on Line 2-3 expressed via the B-matrix is

$$P_{23} = \frac{3[(P_{G2} - D_2) - 2(P_{G3} - D_3)]}{11} = \frac{3(800 - 2P_{G3})}{11}$$

Setting $P_{23} = 100$ MW and solving:

$$\frac{3(800 - 2P_{G3})}{11} = 100 \Rightarrow 800 - 2P_{G3} = \frac{1100}{3} \Rightarrow P_{G3} = \frac{650}{3} \approx 216.7 \text{ MW}$$

$$P_{G1} = 900 - P_{G3} = \frac{2050}{3} \approx 683.3 \text{ MW}$$

$$P_{G1} = \frac{2050}{3} \approx 683.3 \text{ MW}, \quad P_{G2} = 0 \text{ MW}, \quad P_{G3} = \frac{650}{3} \approx 216.7 \text{ MW}$$

Total system cost.

$$TC = \frac{2050}{3} \times 300 + 0 + \frac{650}{3} \times 600 = 205\,000 + 130\,000 = \mathbf{335\,000 \text{ NOK}}$$

Phase angles. The net nodal injections in per unit are

$$P_1 = \frac{2050/3 - 200}{100} = \frac{29}{6} \text{ p.u.}, \quad P_2 = \frac{0 - 200}{100} = -2 \text{ p.u.}, \quad P_3 = \frac{650/3 - 500}{100} = -\frac{17}{6} \text{ p.u.}$$

With $\delta_1 = 0$, the nodal equations are:

$$50\delta_2 - 30\delta_3 = -2 \quad \text{and} \quad -30\delta_2 + 40\delta_3 = -\frac{17}{6}$$

From the first equation: $\delta_2 = -0.04 + 0.6\delta_3$. Substituting into the second:

$$-30(-0.04 + 0.6 \delta_3) + 40 \delta_3 = -\frac{17}{6} \Rightarrow 1.2 - 18 \delta_3 + 40 \delta_3 = -\frac{17}{6} \Rightarrow 22 \delta_3 = -\frac{121}{30}$$

$$\delta_3 = -\frac{11}{60} \approx -0.183 \text{ rad}, \quad \delta_2 = -\frac{3}{20} = -0.150 \text{ rad}$$

$$\delta_1 = 0, \quad \delta_2 = -0.150 \text{ rad}, \quad \delta_3 \approx -0.183 \text{ rad}$$

Power flows.

$$P_{12} = b_{12}(\delta_1 - \delta_2) = 20(0 - (-\frac{3}{20})) = 3 \text{ p.u.} \Rightarrow \mathbf{300 \text{ MW}} \text{ (bus 1} \rightarrow \text{bus 2)}$$

$$P_{13} = b_{13}(\delta_1 - \delta_3) = 10(0 - (-\frac{11}{60})) = \frac{11}{6} \text{ p.u.} \Rightarrow \frac{550}{3} \approx \mathbf{183.3 \text{ MW}} \text{ (bus 1} \rightarrow \text{bus 3)}$$

$$P_{23} = b_{23}(\delta_2 - \delta_3) = 30(-\frac{3}{20} - (-\frac{11}{60})) = 30 \times \frac{1}{30} = 1 \text{ p.u.} \Rightarrow \mathbf{100 \text{ MW}} \text{ (bus 2} \rightarrow \text{bus 3)} \checkmark$$

Nodal balance check:

$$\text{Bus 1: } P_{12} + P_{13} = 300 + \frac{550}{3} = \frac{1450}{3} \text{ MW} = P_{G1} - D_1 \checkmark$$

$$\text{Bus 2: } -P_{12} + P_{23} = -300 + 100 = -200 \text{ MW} = P_{G2} - D_2 \checkmark$$

$$\text{Bus 3: } -P_{13} - P_{23} = -\frac{550}{3} - 100 = -\frac{850}{3} \text{ MW} = P_{G3} - D_3 \checkmark$$

Shadow price of Line 2-3. From the KKT condition for Generator 3:

$$\mu_{23} = \frac{11(c_3 - \lambda)}{6} = \frac{11 \times (600 - 300)}{6} = \mathbf{550 \text{ NOK/MWh}}$$

Nodal prices.

$$\lambda_1 = \lambda - \mu_{23} \cdot 0 = 300 \text{ NOK/MWh}$$

$$\lambda_2 = \lambda - \mu_{23} \cdot \frac{3}{11} = 300 - 550 \times \frac{3}{11} = 300 - 150 = \mathbf{150 \text{ NOK/MWh}}$$

$$\lambda_3 = \lambda - \mu_{23} \cdot \left(-\frac{6}{11}\right) = 300 + 550 \times \frac{6}{11} = 300 + 300 = \mathbf{600 \text{ NOK/MWh}}$$

(e) Reasoning for nodal prices, in particular Node 2

The nodal price at bus 2 is $\lambda_2 = 150 \text{ NOK/MWh}$, which is *lower* than the system energy price $\lambda = 300 \text{ NOK/MWh}$. Three factors explain this:

1. **Bus 2 is on the exporting side of the congested line.** Line 2-3 carries 100 MW *away* from bus 2 toward bus 3. An additional MW of demand at bus 2 reduces the net export from bus 2, which *relaxes* the binding flow constraint on Line 2-3.
2. **Relaxing the constraint allows a cheaper dispatch.** With the constraint slightly relaxed, the system can substitute expensive Generator 3 (600 NOK/MWh) with cheaper Generator 1 (300 NOK/MWh), lowering total cost.
3. **The net effect is a below-system-price LMP at bus 2.** A 1 MW increase in demand at bus 2 causes Generator 1 to increase by 1.5 MW and Generator 3 to decrease by 0.5 MW, giving $\Delta TC = 300 \times 1.5 + 600 \times (-0.5) = 150 \text{ NOK/MWh} = \lambda_2$. Because the counter-flow benefit (saved G3 cost) partly offsets the extra G1 cost, the LMP at bus 2 is below λ_1 .

(f) Effect of increasing demand at Node 2 to 201 MW

The marginal cost of one additional MWh at Node 2 is the nodal price there:

$$\Delta C = \lambda_2 \cdot \Delta P_{D2} = 150 \times 1 = \mathbf{150 \text{ NOK}}$$

The new optimal dispatch (with $P_{G2} = 0$ still optimal) is found from

$$\frac{3(799 - 2P_{G3})}{11} = 100 \Rightarrow P_{G3} = \frac{1297}{6} \approx 216.2 \text{ MW}, \quad P_{G1} = \frac{4109}{6} \approx 684.8 \text{ MW}$$

$$TC_{\text{new}} = \frac{4109}{6} \times 300 + \frac{1297}{6} \times 600 = 205\,450 + 129\,700 = 335\,150 \text{ NOK}$$

$$\Delta TC = 335\,150 - 335\,000 = \mathbf{150 \text{ NOK}}$$

This confirms that $\lambda_2 = 150 \text{ NOK/MWh}$ is indeed the marginal system cost of an extra MW of demand at Node 2.

2 Problem 2: Modeling a 3-node system

This problem analyzes a three-node power system using the DCOPF framework to determine the cost-optimal generation dispatch, power flows, and nodal prices in the network. The system is modeled in Python using Pyomo and solved with the Gurobi optimizer. The results obtained from this model are further used to examine system behavior under different conditions and to evaluate the economic and environmental implications of various policy mechanisms.

2.1 Task 2-1: Defining the Mathematical Formulation for DCOPF

2.1 a) Objective of the optimization problem

The objective is to determine the cost-optimal dispatch of electricity generation in the system. This is achieved by minimizing the total generation cost:

$$\min \sum_{g \in G} c_g P_g \quad (1)$$

where c_g [NOK/MWh] is the marginal generation cost and P_g [MW] is the power output of generator g , and G denotes the set of generators in the system. This is the standard DCOPF objective, which uses a linearized representation of the AC power flow equations to model the transmission network (Farahmand, 2026c).

2.1 b) Constraints regarding power generation

Each generator's output must remain within its technical operating range:

$$P_g^{\min} \leq P_g \leq P_g^{\max} \quad \forall g \in G \quad (2)$$

where $P_g^{\min}$ and $P_g^{\max}$ denote the minimum and maximum generation capacities of generator g (Farahmand, 2026c). These constraints ensure that the generation dispatch remains within feasible operating limits.

2.1 c) Constraints regarding power flow

The DCOPF model is based on the DC power flow approximation, which provides a linear representation of power flows in the transmission network. Under the DC power flow approximation, voltage magnitudes are assumed to be 1 p.u., resistances are neglected, and angle

differences are small. The power flow on a transmission line between bus n and bus m then becomes a linear function of the voltage angle difference (Farahmand, 2026c):

$$F_{nm} = B_{nm}(\theta_n - \theta_m) \quad (3)$$

where F_{nm} [MW] is the power flow from bus n to bus m , B_{nm} is the line susceptance, and θ_n , θ_m are the voltage phase angles at the respective buses (Farahmand, 2026c). Each transmission line is also subject to thermal capacity limits in both directions:

$$-F_{nm}^{\max} \leq F_{nm} \leq F_{nm}^{\max} \quad (4)$$

where $F_{nm}^{\max}$ denotes the maximum allowable power flow on the line. Equivalently, substituting the DC flow expression gives $-F_{nm}^{\max} \leq B_{nm}(\theta_n - \theta_m) \leq F_{nm}^{\max}$ (Farahmand, 2026c). These constraints ensure that flows remain within the physical limits of each transmission line.

2.1 d) Complete DCOPF formulation

Combining the objective with the system constraints, the DCOPF problem can be formulated as:

$$\min \quad \sum_{g \in G} c_g P_g \quad (5a)$$

$$\text{subject to: } \sum_{g \in G_n} P_g - D_n = \sum_{m \in \Omega_n} B_{nm}(\theta_n - \theta_m) \quad \forall n \in N \quad (\lambda_n) \quad (5b)$$

$$-F_{nm}^{\max} \leq B_{nm}(\theta_n - \theta_m) \leq F_{nm}^{\max} \quad \forall (n, m) \in L \quad (5c)$$

$$P_g^{\min} \leq P_g \leq P_g^{\max} \quad \forall g \in G \quad (5d)$$

$$\theta_{\text{ref}} = 0 \quad (5e)$$

The decision variables are the generator outputs P_g and the voltage angles θ_n . The nodal power balance constraint ensures that, at each bus n , total generation minus demand D_n equals the net power flow to connected buses, where G_n denotes the set of generators at bus n and Ω_n denotes the set of buses connected to n (Farahmand, 2026c). The dual variable λ_n represents the nodal price (Farahmand, 2026b). The reference bus constraint fixes one voltage angle (typically set to zero) to ensure a unique solution, since only angle differences affect power flows (Farahmand, 2026c).

2.2 Task 2-2: Creating the Optimization Problem

2.2 a) DCOPF solution and system analysis

The DCOPF problem defined in Task 2-1 was implemented in Python using Pyomo and solved with the Gurobi optimizer. The input data was read from the provided Excel file, using a per-unit base of 1000 MVA.

The optimal generation dispatch is presented in Table 2.1. Generator 1 produces 683.33 MW and Generator 3 produces 216.67 MW, while Generator 2 is not dispatched because its marginal cost (1000 NOK/MWh) exceeds what the system needs to pay given the available network capacity. The total system cost is 335 000 NOK.

Table 2.1: Optimal generation dispatch in the DCOPF solution.

Generator	Node	Cost [NOK/MWh]	Output [MW]	Status
Gen 1	1	300	683.33	Partial
Gen 2	2	1000	0	Not dispatched
Gen 3	3	600	216.67	Partial

The resulting transmission flows are shown in Table 2.2. The transmission line between Node 2 and Node 3 reaches its maximum capacity of 100 MW and is the only binding constraint in the system. No generator capacity constraint is active at the optimum.

Table 2.2: Power flows on transmission lines.

Line	Flow [MW]	Capacity [MW]	Status
Line 1–2	300.00	500	Non-binding
Line 1–3	183.33	500	Non-binding
Line 2–3	100.00	100	Binding

The shadow price of the binding line constraint is $\mu_{2-3} = -550$ NOK/MW. Since the problem is a minimization, this means that increasing the line capacity by 1 MW would reduce the total system cost by 550 NOK.

The nodal prices are presented in Table 2.3. These correspond to the dual variables of the power balance constraints (5b), representing the marginal cost of serving one additional MW of demand at each bus (Farahmand, 2026b). The price differences across the system reflect congestion caused by the binding constraint on line 2–3.

Table 2.3: Nodal prices from the DCOPF solution.

Node	Price [NOK/MWh]
1	300
2	150
3	600

2.2 b) Updated nodal prices

The marginal cost of Generator 3 is increased to 1000 NOK/MWh and the model is re-solved. The generation dispatch remains identical to Task 2-2a, since the binding line 2–3 constraint forces the same physical power flows regardless of the cost parameter. The updated nodal prices are presented in Table 2.4.

Table 2.4: Updated nodal prices after increasing the marginal cost at Node 3

Node	Price [NOK/MWh]
1	300
2	–50
3	1000

Node 1 retains a price of 300 NOK/MWh, corresponding to the marginal cost of Generator 1. The price at Node 3 increases to 1000 NOK/MWh because the congestion on line 2–3 prevents sufficient power from being imported, forcing the expensive local generator to set the price (Farahmand, 2026b). At Node 2, the nodal price becomes negative (–50 NOK/MWh). This occurs because, given the binding line 2–3 constraint, an additional MW of demand at Node 2 would allow more power to flow towards Node 3 via the constrained line, displacing expensive local generation there and reducing the total system cost.

2.2 c) Flow-based market coupling and comparison with DCOPF

The DCOPF formulation models the network at the nodal level, with voltage phase angles θ_n as explicit decision variables. Because flows are determined by angle differences across the full network, a change in generation at one node automatically affects flows on all connected lines (Farahmand, 2026c).

In the FBMC approach, voltage angles are eliminated. The network is instead represented through Power Transfer Distribution Factors (PTDFs), which are pre-computed from the network admittance matrix (Farahmand, 2026a). The flow on a critical network element l is ex-

pressed as a linear function of zonal net positions:

$$F_l = \sum_z \text{PTDF}_{l,z} \cdot \text{NP}_z \quad (6)$$

where $\text{PTDF}_{l,z}$ is the sensitivity of line l to a unit injection in zone z , and NP_z is the net position of zone z . The transmission constraints become:

$$-F_l^{\max} \leq \sum_z \text{PTDF}_{l,z} \cdot \text{NP}_z \leq F_l^{\max} \quad (7)$$

This makes FBMC more tractable for large multi-zone market clearing, since the optimisation does not require voltage angles, but it relies on the accuracy of PTDFs computed offline by the system operator (Farahmand, 2026a, 2026c). The main differences are summarised in Table 2.5.

Table 2.5: Comparison of the DCOPF and FBMC approaches.

Aspect	DCOPF	FBMC
Network representation	Nodal	Zonal
Decision variables	θ_n, P_g	NP_z, P_g
Flow calculation	$F_{nm} = B_{nm}(\theta_n - \theta_m)$	$F_l = \sum_z \text{PTDF}_{l,z} \cdot \text{NP}_z$
Parallel path flows	Captured directly by angle equations	Approximated via PTDFs
Constraint basis	Line susceptances	Pre-computed PTDFs

2.2 d) Implementation of the FBMC model

The FBMC model was implemented in Python using Pyomo and solved with the Gurobi optimizer. The key step is computing the PTDF matrix, which describes how a unit net injection at node n (with simultaneous withdrawal at the reference node) affects the flow on each line (Farahmand, 2026c). The PTDFs are computed as $\mathbf{H} = \tilde{\mathbf{B}}_f \tilde{\mathbf{B}}_{\text{bus}}^{-1}$, where $\tilde{\mathbf{B}}_{\text{bus}}$ is the bus susceptance matrix with the reference node row and column removed, and $\tilde{\mathbf{B}}_f$ is the branch-bus connection matrix without the reference node column (Farahmand, 2026c). For this system:

$$\tilde{\mathbf{B}}_{\text{bus}} = \begin{bmatrix} 50 & -30 \\ -30 & 40 \end{bmatrix}, \quad \tilde{\mathbf{B}}_{\text{bus}}^{-1} = \frac{1}{1100} \begin{bmatrix} 40 & 30 \\ 30 & 50 \end{bmatrix} \quad (8)$$

The resulting PTDF matrix is shown in Table 2.6.

Table 2.6: PTDF matrix for the 3-node system (reference: Node 1).

Line	Node 2	Node 3
Line 1-2	-0.7273	-0.5455
Line 1-3	-0.2727	-0.4545
Line 2-3	+0.2727	-0.5455

The FBMC model minimises total generation cost subject to a global power balance and PTDF-based line flow constraints:

$$-F_l^{\max} \leq \sum_{n \in \{2,3\}} \text{PTDF}_{l,n} \cdot \text{NP}_n \leq F_l^{\max} \quad \forall l \quad (9)$$

where $\text{NP}_n = P_{\text{gen},n} - D_n$ is the net position of node n . The results are compared with the DCOPF solution in Table 2.7.

Table 2.7: Comparison of DCOPF and FBMC results.

	DCOPF	FBMC
P_1 (MW)	683.33	683.33
P_2 (MW)	0	0
P_3 (MW)	216.67	216.67
Total cost (NOK)	335 000	335 000
λ_1 (NOK/MWh)	300	300
λ_2 (NOK/MWh)	150	150
λ_3 (NOK/MWh)	600	600
Line 2-3 flow (MW)	100	100

The dispatch, flows, prices, and total cost are identical. In the FBMC model, the nodal prices are recovered from the dual variables: the global balance constraint gives the reference price $\lambda_1 = \lambda_{\text{sys}} = 300$ NOK/MWh, and prices at the remaining nodes follow from $\lambda_n = \lambda_{\text{sys}} + \mu_{2-3} \cdot \text{PTDF}_{2-3,n}$, where $\mu_{2-3} = -550$ NOK/MW is the shadow price of the binding line constraint. This yields $\lambda_2 = 300 + (-550) \cdot 3/11 = 150$ NOK/MWh and $\lambda_3 = 300 + (-550) \cdot (-6/11) = 600$ NOK/MWh, consistent with the DCOPF results.

The identical results follow from the structure of this system: each node constitutes its own bidding zone with a single generator, making the PTDF-based flow constraints mathematically equivalent to the DC power flow equations. The two formulations differ only in representation, DCOPF uses voltage angles and per-node balance, while FBMC uses zonal net positions and pre-computed PTDFs, but produce the same solution when zones contain a single node.

2.3 Task 2-3: Extending the Model: Multiple Generators

Node 1 is extended to include three generators with different marginal costs and production capacities: $\text{Gen}_{1,1}$ (200 NOK/MWh, 300 MW), $\text{Gen}_{1,2}$ (300 NOK/MWh, 400 MW), and $\text{Gen}_{1,3}$ (800 NOK/MWh, 300 MW). All other system parameters remain identical to those in Task 2-2.

The optimal dispatch is presented in Table 2.8 and Figure 2.1. The optimizer exhausts the cheapest generation first: $\text{Gen}_{1,1}$ is fully utilised at its 300 MW capacity limit, while $\text{Gen}_{1,2}$ supplies the remaining demand at Node 1 with 383.33 MW. The more expensive $\text{Gen}_{1,3}$ (800 NOK/MWh) is not dispatched, as its cost exceeds the nodal price. Gen_2 at Node 2 is also not dispatched, and Gen_3 produces 216.67 MW at Node 3.

Table 2.8: Optimal generation dispatch for Task 2-3.

Generator	Node	Cost [NOK/MWh]	Cap [MW]	Dispatch [MW]
$\text{Gen}_{1,1}$	1	200	300	300.00
$\text{Gen}_{1,2}$	1	300	400	383.33
$\text{Gen}_{1,3}$	1	800	300	0
Gen_2	2	1000	1000	0
Gen_3	3	600	1000	216.67

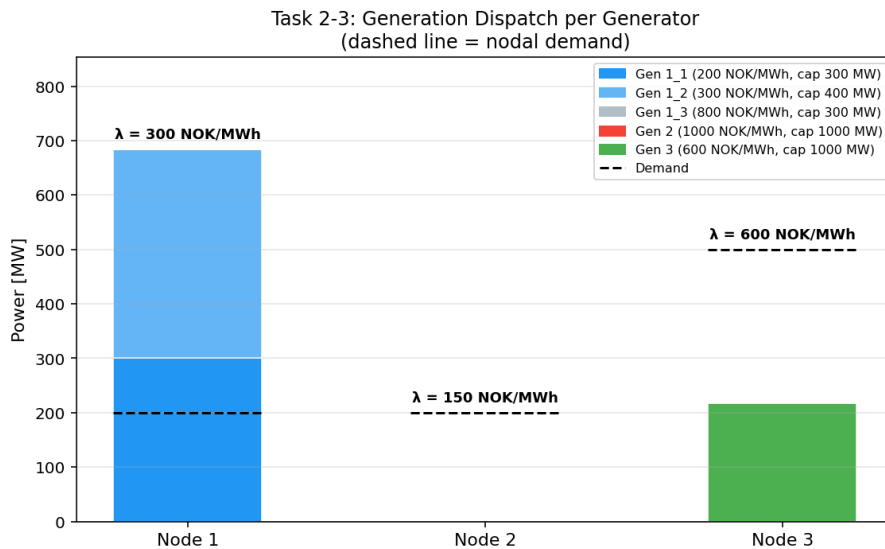


Figure 2.1: Generation dispatch per generator. Dashed lines indicate nodal demand, and nodal prices are annotated above each bar.

The resulting nodal prices ($\lambda_1 = 300$, $\lambda_2 = 150$, $\lambda_3 = 600$ NOK/MWh) and line flows ($F_{12} = 300$ MW, $F_{13} = 183.33$ MW, $F_{23} = 100$ MW) are identical to those obtained in Task 2-2a. This is expected since the total net injection at each node is unchanged, so the power flows and congestion pattern in the network remain the same. Line 2-3 is the only binding transmission

constraint, with a shadow price of -550 NOK/MW.

$\text{Gen}_{1,1}$ is also binding at its capacity limit of 300 MW, with a shadow price of -100 NOK/MW. This equals the difference between the nodal price at Node 1 (300 NOK/MWh) and the generator's marginal cost (200 NOK/MWh), indicating that each additional MW of capacity at $\text{Gen}_{1,1}$ would reduce the total system cost by 100 NOK by displacing more expensive generation from $\text{Gen}_{1,2}$.

The total system cost is 305 000 NOK, a reduction of 30 000 NOK compared to Task 2-2. This occurs because 300 MW of generation is now supplied by $\text{Gen}_{1,1}$ at 200 NOK/MWh instead of the single generator at 300 NOK/MWh in the base case.

2.4 Task 2-4: Extending the Model: Multiple Loads

2.4 a) Inflexible loads

Node 2 is extended to include three separate loads: $\text{Load}_{2,1}$ (200 MW), $\text{Load}_{2,2}$ (250 MW), and $\text{Load}_{2,3}$ (250 MW), all treated as inflexible in this sub-task. Total nodal demand becomes $D_1 = 200$ MW, $D_2 = 700$ MW, and $D_3 = 500$ MW, giving a system total of 1400 MW.

The optimal dispatch is shown in Table 2.9 and Figure 2.2. The large increase in demand at Node 2 forces major changes relative to Task 2-3. $\text{Gen}_{1,1}$ and $\text{Gen}_{1,2}$ are both fully dispatched at their capacity limits. $\text{Gen}_{1,3}$ (800 NOK/MWh) is now partially dispatched at 216.67 MW, and Gen_2 (1000 NOK/MWh) at Node 2 produces 100 MW locally. Gen_3 covers 383.33 MW at Node 3.

Table 2.9: Optimal generation dispatch for Task 2-4a.

Generator	Node	Cost [NOK/MWh]	Cap [MW]	Dispatch [MW]	Status
$\text{Gen}_{1,1}$	1	200	300	300.00	At capacity
$\text{Gen}_{1,2}$	1	300	400	400.00	At capacity
$\text{Gen}_{1,3}$	1	800	300	216.67	Partial
Gen_2	2	1000	1000	100.00	Partial
Gen_3	3	600	1000	383.33	Partial

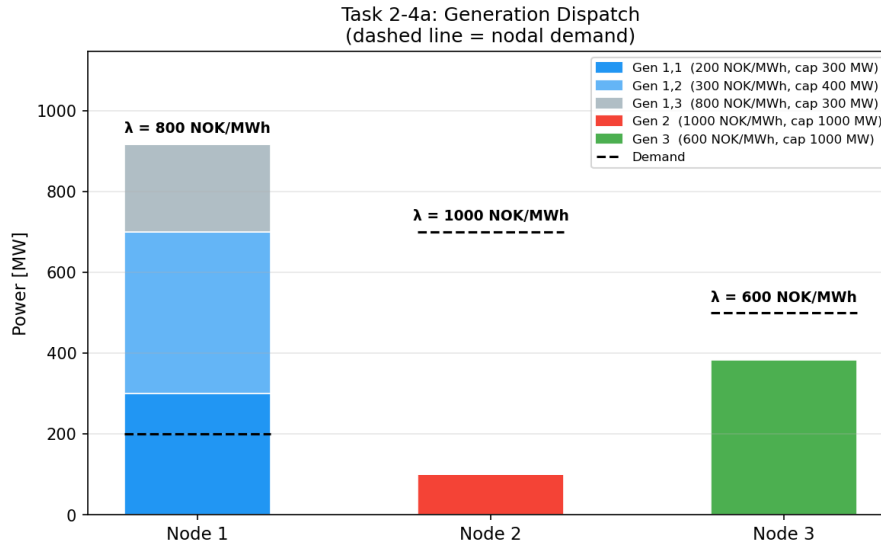


Figure 2.2: Generation dispatch per generator for Task 2-4a. Dashed lines indicate nodal demand. Nodal prices are annotated above each bar.

The line flows are shown in Table 2.10. Two transmission constraints are now binding: Line 1–2 reaches its upper limit of 500 MW (reflecting the large net export from Node 1 towards Node 2), and Line 2–3 is binding in the negative direction at -100 MW (power flows from Node 3 towards Node 2 at full capacity).

Table 2.10: Transmission line flows for Task 2-4a.

Line	Flow [MW]	Capacity [MW]	Status
Line 1–2	+500.00	500	Binding (upper)
Line 1–3	+216.67	500	Non-binding
Line 2–3	–100.00	100	Binding (lower)

The nodal prices are $\lambda_1 = 800$, $\lambda_2 = 1000$, and $\lambda_3 = 600$ NOK/MWh. Each price equals the marginal cost of the partially dispatched generator at that node: $\text{Gen}_{1,3}$ at Node 1, Gen_2 at Node 2, and Gen_3 at Node 3. The price at Node 2 is the highest in the system because both connecting lines are fully utilised, forcing expensive local generation to set the price.

Four constraints are binding. $\text{Gen}_{1,1}$ and $\text{Gen}_{1,2}$ have shadow prices of -600 and -500 NOK/MW, equal to the nodal price (800) minus their marginal costs (200 and 300). Line 1–2 has a shadow price of -100 NOK/MW (upper bound), and Line 2–3 has a shadow price of $+466.67$ NOK/MW (lower bound). The positive shadow price on the lower bound reflects the value of increased import capacity from Node 3 to Node 2: each additional MW would reduce total system cost by 466.67 NOK.

The total system cost is 683 333 NOK, a substantial increase from 305 000 NOK in Task 2-3,

driven by the large demand increase at Node 2 requiring the dispatch of expensive generators Gen_{1,3} and Gen₂.

2.4 b) Social welfare maximization and change in objective function

The objective changes from minimising total generation cost to maximising social welfare, defined as total consumer utility minus total generation cost:

$$\max \sum_{d \in D_{\text{flex}}} \text{WTP}_d \cdot P_d - \sum_{g \in G} c_g \cdot P_g \quad (10)$$

where WTP_d is the marginal willingness-to-pay of flexible load d and P_d is the quantity served. The flexible loads at Node 2 (Load_{2,1}, Load_{2,2}, Load_{2,3}) are bounded by their maximum capacity: $0 \leq P_d \leq D_d^{\text{max}}$.

Load₁ and Load₃ have no given WTP and are assumed to be inflexible, meaning that they represent essential demand that must always be fully served. This is justified by the assumption that their WTP is effectively infinite, meaning no price would cause them to reduce consumption. They are therefore modelled as fixed equality constraints ($P_d = D_d^{\text{max}}$), ensuring full coverage regardless of the market outcome.

2.4 c) Social welfare maximization results

The social welfare maximization model is solved with all generators from Task 2-3 and all loads. Results are shown in Tables 2.11–2.12 and Figure 2.3.

Gen_{1,1} and Gen_{1,2} are fully dispatched at their capacity limits, and Gen₃ produces 450 MW. Gen_{1,3} and Gen₂ are not dispatched. No transmission line is binding (flows: $F_{12} = 354.55$ MW, $F_{13} = 145.45$ MW, $F_{23} = -95.45$ MW).

Table 2.11: Generation dispatch for Task 2-4c.

Generator	Node	Cost [NOK/MWh]	Cap [MW]	Dispatch [MW]
Gen _{1,1}	1	200	300	300.00
Gen _{1,2}	1	300	400	400.00
Gen _{1,3}	1	800	300	0
Gen ₂	2	1000	1000	0
Gen ₃	3	600	1000	450.00

Table 2.12: Load served for Task 2-4c.

Load	Node	WTP [NOK/MWh]	Max [MW]	Served [MW]	Status
Load ₁	1	–	200	200.00	Inflexible
Load _{2,1}	2	1300	200	200.00	Fully served
Load _{2,2}	2	800	250	250.00	Fully served
Load _{2,3}	2	500	250	0	Curtailed
Load ₃	3	–	500	500.00	Inflexible

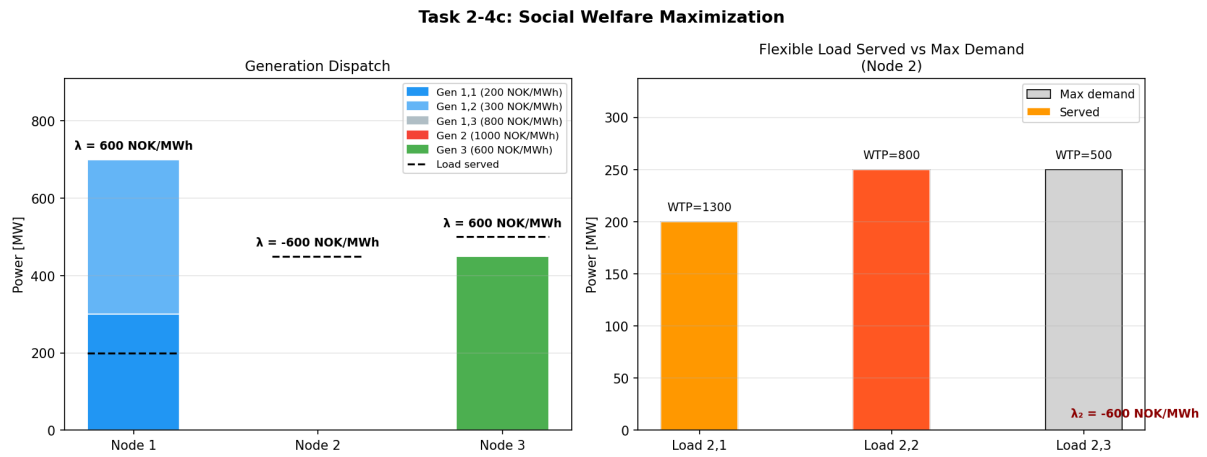


Figure 2.3: Left: generation dispatch per node with nodal prices annotated. Right: flexible load served vs maximum demand at Node 2.

Load_{2,3} (WTP = 500 NOK/MWh) is curtailed because its willingness-to-pay is below the nodal price at Node 2 (600 NOK/MWh). Load_{2,1} and Load_{2,2} are fully served since their WTP exceeds the nodal price.

All nodal prices equal 600 NOK/MWh. The system is uncongested, so prices equalise across all nodes and are set by the marginal generator, Gen₃ (cost = 600 NOK/MWh). The curtailment of Load_{2,3} follows directly: serving it would cost 600 NOK/MWh but generate only 500 NOK/MWh in consumer utility, reducing social welfare.

Four constraints are binding. Gen_{1,1} and Gen_{1,2} are at capacity with shadow prices of +400 and +300 NOK/MW, equal to the nodal price minus each generator's marginal cost (600 – 200 and 600 – 300), representing the gain in social welfare from relaxing each capacity limit by 1 MW. Load_{2,1} and Load_{2,2} are fully served with shadow prices of +700 and +200 NOK/MW, representing the consumer surplus at each load (WTP minus nodal price: 1300 – 600 and 800 – 600). The total social welfare is 10 000 NOK (consumer utility 460 000 NOK minus generation cost 450 000 NOK).

2.5 Task 2-5: Extending the Model: Environmental Constraints

2.5 A) Modeling the Clean Energy Standard

A Clean Energy Standard (CES) mandates that a minimum share of total electricity generation must come from zero-carbon sources (Farahmand, 2026d). It is modelled by adding a single inequality constraint to the social welfare maximization model from Task 2-4c:

$$\sum_{g \in G_{\text{clean}}} P_g \geq \alpha \sum_{g \in G} P_g \quad (11)$$

where $\alpha = 0.20$ is the required clean share and G_{clean} is the set of generators with zero CO₂ emissions. Based on the emission data, only Gen₂ (0 kg/MWh) qualifies as clean, while all other generators carry positive emissions.

2.5 B) Results with the CES constraint

The social welfare maximization model from Task 2-4c is extended with the 20% CES constraint. The results are compared to the baseline (no CES) in Table 2.13.

Table 2.13: Comparison of results without and with the 20% CES.

	No CES (Task 2-4c)	20% CES
Gen _{1,1} [MW]	300.00	300.00
Gen _{1,2} [MW]	400.00	400.00
Gen _{1,3} [MW]	0	0
Gen ₂ [MW]	0	230.00
Gen ₃ [MW]	450.00	220.00
λ [NOK/MWh]	600	680
Social welfare [NOK]	10 000	−82 000
CO ₂ [tonnes]	1 180	950

The CES forces Gen₂, the only zero-emission generator, to produce 230 MW, exactly meeting the 20% requirement ($230/1150 = 20\%$). To accommodate this, Gen₃ is reduced from 450 MW to 220 MW, while total generation remains at 1150 MW (load served is unchanged, with Load_{2,3} still curtailed). No transmission line becomes binding.

The nodal price rises from 600 to 680 NOK/MWh at all nodes. The system remains uncongested, so a single system price prevails. The price increase can be understood through the KKT stationarity conditions. With the CES dual value $\kappa = 400$ NOK/MW, the nodal price is:

$$\lambda = c_{\text{Gen}_3} + \kappa \cdot \alpha = 600 + 400 \times 0.20 = 680 \text{ NOK/MWh} \quad (12)$$

This reflects the fact that the CES effectively raises the implicit cost of the marginal non-clean generator by $\kappa \cdot \alpha$, since each additional MWh from a non-clean source tightens the constraint. Equivalently, from the clean side: $\lambda = c_{\text{Gen}_2} - \kappa(1 - \alpha) = 1000 - 400 \times 0.80 = 680$ NOK/MWh. Social welfare falls from 10 000 to -82 000 NOK, as the policy forces costly generation. Total CO₂ emissions fall from 1 180 to 950 tonnes.

2.5 C) Dual value of the CES constraint

The dual value of the CES constraint is $\kappa = 400$ NOK/MW. It represents the marginal cost of the CES policy: tightening the clean generation requirement by 1 MW would reduce social welfare by 400 NOK. Equivalently, it is the implicit green certificate price, the subsidy per MWh needed to make Gen₂ competitive with the marginal non-clean generator at the current dispatch (Farahmand, 2026d). The units are NOK/MW (or equivalently NOK/MWh for a one-hour model), reflecting the change in the objective function per unit relaxation of the clean generation requirement.

2.5 a) Modeling the emission cap

A cap-and-trade system limits total CO₂ emissions by imposing an upper bound on aggregate emissions from all generators (Farahmand, 2026d). The CES constraint from equation (11) is replaced by a single emission cap constraint:

$$\sum_{g \in G} E_g^{\text{CO}_2} \cdot P_g \leq \bar{E}^{\text{CO}_2} \quad (13)$$

where $E_g^{\text{CO}_2}$ is the emission intensity of generator g in kg/MWh, and $\bar{E}^{\text{CO}_2}$ is the emission cap. As specified in the task, the cap is set equal to the system-wide emissions obtained under the CES policy: $\bar{E}^{\text{CO}_2} = 950\,000$ kg. Unlike the CES, which mandates a clean generation share, cap-and-trade directly constrains total emissions regardless of which generators produce the power.

2.5 b) Results and dual interpretation

The social welfare maximization model is extended with the emission cap, replacing the CES constraint. The results are presented in Table 2.14.

Table 2.14: Comparison of results: no policy, CES, and cap-and-trade.

	No policy	CES	Cap-and-trade
Gen _{1,1} [MW]	300	300	300
Gen _{1,2} [MW]	400	400	400
Gen _{1,3} [MW]	0	0	0
Gen ₂ [MW]	0	230	0
Gen ₃ [MW]	450	220	220
Load _{2,2} [MW]	250	250	20
λ [NOK/MWh]	600	680	800
Gen. cost [NOK]	450 000	542 000	312 000
Social welfare [NOK]	10 000	−82 000	−36 000
CO ₂ [tonnes]	1 180	950	950

Under cap-and-trade, the optimizer reduces emissions by curtailing demand rather than dispatching expensive clean generation. Gen₂ remains undispached because its marginal cost of 1000 NOK/MWh exceeds the nodal price, even though it has zero emissions. Instead, Gen₃ is reduced from 450 to 220 MW and Load_{2,2} is curtailed from 250 to 20 MW, bringing total generation and demand down to 920 MW.

The system remains uncongested, and all nodal prices equal 800 NOK/MWh. The price is set by the effective marginal cost of Gen₃, which now includes the emission cost:

$$\lambda = c_{\text{Gen}_3} + \kappa \cdot E_{\text{Gen}_3}^{\text{CO}_2} = 600 + 0.2 \times 1000 = 800 \text{ NOK/MWh} \quad (14)$$

The dual value of the emission cap constraint is $\kappa = 0.2$ NOK/kg (equivalently 200 NOK/tonne). It represents the CO₂ price: the marginal reduction in social welfare from tightening the emission cap by one kilogram. Equivalently, κ is the market-clearing price of an emission allowance in the cap-and-trade system (Farahmand, 2026d).

2.5 c) Comparison of policy results

Table 2.15 extends the comparison to include transmission flows.

Table 2.15: Full comparison including transmission flows.

	No policy	CES	Cap-and-trade
F_{12} [MW]	354.55	312.73	312.73
F_{13} [MW]	145.45	187.27	187.27
F_{23} [MW]	−95.45	92.73	92.73

The two policies achieve the same emission reduction (from 1180 to 950 tonnes) through different mechanisms. The CES forces Gen₂ into dispatch at 230 MW to satisfy the 20% clean share, while total generation remains at 1150 MW. Cap-and-trade does not prescribe which technology must run; the optimizer instead reduces Gen₃ and curtails Load_{2,2}, bringing total output down to 920 MW. Gen₂ remains undischarged because its marginal cost exceeds the nodal price even with zero emission cost.

Cap-and-trade produces higher nodal prices (800 vs. 680 NOK/MWh) despite achieving higher social welfare (−36000 vs. −82000 NOK). Under the CES, the certificate cost is spread over the clean share ($\lambda = c_{\text{Gen}_3} + \kappa_{\text{CES}} \cdot \alpha = 600 + 400 \times 0.20 = 680$), whereas under cap-and-trade the full CO₂ cost of the marginal generator is internalised ($\lambda = c_{\text{Gen}_3} + \kappa_{\text{cap}} \cdot E_{\text{Gen}_3} = 600 + 0.2 \times 1000 = 800$). The higher price signal drives the demand-side curtailment that makes cap-and-trade more cost-effective.

The CES and cap-and-trade produce identical line flows despite different dispatch patterns. This occurs because both policies yield the same net injection at each node: at Node 2, the CES injects 230 MW of generation while serving 450 MW of load (net −220 MW), whereas cap-and-trade injects 0 MW while serving only 220 MW (also net −220 MW). No transmission line becomes binding in any of the three scenarios.

2.5 d) Pros and cons of each policy

Both policies achieve the same emission reduction, but cap-and-trade is the more cost-effective alternative, with social welfare of −36000 NOK compared to −82000 NOK under the CES. This difference arises because cap-and-trade only constrains total emissions and lets the market find the cheapest way to comply — in this case, reducing low-value demand (Load_{2,2}, WTP = 800 NOK/MWh) rather than dispatching the expensive clean generator (Gen₂, 1000 NOK/MWh). The CES cannot use this flexibility because it prescribes a technology requirement rather than an emission outcome (Farahmand, 2026d).

However, the CES has advantages that the welfare numbers alone do not capture. By guaranteeing a market share for zero-carbon generators, the CES provides long-term investment signals for clean capacity, which is valuable when innovation spillovers and learning-by-doing effects are present (Farahmand, 2026d). The CES also maintains total served demand at 1150 MW, whereas cap-and-trade reduces it to 920 MW and produces higher electricity prices (800 vs. 680 NOK/MWh), increasing costs for all consumers including those with inflexible demand.

In summary, cap-and-trade is preferable when the objective is purely cost-effective emission reduction, while the CES is better suited when the policy also aims to promote long-term investment in clean generation capacity.

3 Problem 3: Nordic System FBMC

3.1 Task 3-1: Getting Familiar With the Model

a) **Approximation methods.** The model supports two clearing methods, selected via the `DCFlow` parameter in the Declarations sheet (1 = FBMC, 0 = ATC), read at line 63. Under FBMC (line 244), AC flows are computed from PTDF sensitivities applied to zonal net injections, with a single global balance constraint. Under ATC (line 260), per-node balance is enforced via the bus-incidence matrix, and each AC flow is a free decision variable bounded only by its bilateral capacity limit.

Reference node. Node 6 (SE1) is the reference, declared in row 2 of Declarations and passed to `Calculate_PTDF()` (line 134), where its row and column are removed from **B** before inversion. Fixing $\delta_{\text{ref}} = 0$ anchors the angle frame. All PTDF entries then express line-flow sensitivity to a unit injection at zone n relative to SE1. The choice is arbitrary, because flows depend only on angle *differences*, dispatch, flows, and prices are invariant to it.

Objective function. Defined at lines 215-216, the model minimises total system cost:

$$\min \sum_{n \in \mathcal{N}} c_n P_n^{\text{gen}} + c^{\text{shed}} \sum_{n \in \mathcal{N}} P_n^{\text{shed}} \quad (15)$$

where c_n [€/MWh] is the marginal generation cost, P_n^{gen} [MW] the dispatch, $c^{\text{shed}} = 3200$ €/MWh the curtailment penalty, and P_n^{shed} [MW] the load shed at zone n .

Insufficient generation capacity. If available capacity cannot meet demand, the load-shedding variable `shed[n]` (line 209) enters both the power-balance constraint and the objective at 3200 €/MWh. Since this exceeds all generation costs, the optimiser exhausts every available generator before curtailing load. When shedding is nonetheless forced, the dual price at the affected zone rises to 3200 €/MWh, signalling local scarcity.

Required Excel filename. The script is hardcoded to read `Nordic_wet.xlsx` from the working directory. To run the dry-year scenario, we need to change this to `Nordic_dry.xlsx`.

b) Both formulations minimise total system cost:

$$\min \sum_{n \in \mathcal{N}} c_n P_n^{\text{gen}} + c^{\text{shed}} \sum_{n \in \mathcal{N}} P_n^{\text{shed}} \quad (16)$$

where c_n is the marginal generation cost at zone n and $c^{\text{shed}} = 3200$ €/MWh is the curtailment penalty. Both share generator capacity, HVDC, and non-negativity constraints:

$$P_n^{\min} \leq P_n^{\text{gen}} \leq P_n^{\max} \quad \forall n \in \mathcal{N} \quad (17)$$

$$-F_h^{\text{cap}} \leq f_h^{\text{DC}} \leq F_h^{\text{cap}} \quad \forall h \in \mathcal{H} \quad (18)$$

$$P_n^{\text{shed}} \geq 0 \quad \forall n \in \mathcal{N} \quad (19)$$

They differ only in how AC transmission is modelled.

FBMC. Zonal net injections,

$$\text{INJ}_n = P_n^{\text{gen}} - D_n + P_n^{\text{shed}} - \sum_{h \in \mathcal{H}} A_{h,n}^{\text{DC}} \cdot f_h^{\text{DC}}, \quad (20)$$

must satisfy $\sum_n \text{INJ}_n = 0$. AC flows follow from the PTDF matrix:

$$f_l^{\text{AC}} = \sum_{n \in \mathcal{N}} \text{PTDF}_{l,n} \cdot \text{INJ}_n \quad \forall l \in \mathcal{L}, \quad (21)$$

where $\text{PTDF}_{l,n}$ is computed from the inverse of the reduced nodal susceptance matrix with SE1 as reference. Flows are bounded by asymmetric thermal limits:

$$-F_l^{\text{to}} \leq f_l^{\text{AC}} \leq F_l^{\text{from}} \quad \forall l \in \mathcal{L}. \quad (22)$$

ATC. A per-node balance replaces PTDF physics:

$$P_n^{\text{gen}} + P_n^{\text{shed}} = D_n + \sum_{l \in \mathcal{L}} X_{l,n} \cdot f_l^{\text{AC}} + \sum_{h \in \mathcal{H}} A_{h,n}^{\text{DC}} \cdot f_h^{\text{DC}} \quad \forall n \in \mathcal{N}, \quad (23)$$

where $X_{l,n} \in \{+1, -1, 0\}$ encodes line orientation and each AC flow is a free variable bounded only by (22).

Because FBMC couples every AC flow to the full injection pattern via (21), any generation change propagates across the entire network. ATC instead treats flows as independent, ignoring loop flows entirely — which can produce physically infeasible dispatches in meshed networks. HVDC links are identical in both: controllable point-to-point flows independent of network impedance.

3.2 Task 3-2: Analyzing a Wet-Year Scenario

In this scenario, abundant hydropower inflow gives Norwegian and Swedish zones high generation capacities relative to demand, creating exportable surplus that must be transported through the transmission network. All results are obtained using the GLPK solver.

a) FBMC is the more physically realistic representation. PTDF sensitivities distribute a single zonal injection across all parallel paths according to electrical impedance, exactly as in a real meshed AC grid, so loop flows are captured endogenously. ATC treats each line as an independent bilateral pipe, with flows bounded only by transfer capacities and no coupling between corridors. The optimiser may therefore schedule a dispatch that would physically overload a line not party to the transaction.

Effect on the objective value. FBMC costs 3 292 168 € against 3 265 108 € under ATC, a gap of 27 060 € (0.8%). ATC is cheaper precisely because it is less constrained. Without PTDF coupling, cheaper generators dispatch more freely. That solution may nonetheless be physically infeasible, applying DC power flow to the ATC dispatch could reveal overloaded lines. The FBMC premium therefore reflects the true cost of respecting network physics, and the gap quantifies the price of loop flows.

b) i) **Total system cost.** Table 3.1 summarises the objective values.

Table 3.1: Total system cost comparison (wet year).

Method	Total Cost [€]	Difference
FBMC	3 292 168	—
ATC	3 265 108	−27 060 (−0.8 %)

ii) **Generation and nodal prices.** Table A.1 and Figure A.1 report generation and prices for both methods. Three observations stand out.

1. **Cheap hydro zones saturate under both methods.** NO4 (27 €/MWh), NO3 (38 €/MWh), SE1 and SE2 (35 €/MWh) are fully dispatched regardless of method.
2. **The main dispatch difference is at NO5 and NO2.** ATC dispatches NO5 (63 €/MWh) at its full 6500 MW. FBMC limits it to 4949 MW because the NO5→NO2 corridor (600 MW) binds under PTDF physics, restricting how much cheap NO5 power reaches deficit zones. The shortfall shifts to the costlier NO2 (77 €/MWh), which rises from 5912 MW to 8227 MW. ATC avoids this by routing 3900 MW on NO5→NO1 without accounting for parallel-path impacts.
3. **SE4 output drops under FBMC.** Once loop flows are accounted for, cheaper imports

via SE3→SE4 serve SE4 demand more efficiently, reducing local generation from 1600 MW (ATC) to 836 MW.

Under ATC, the absence of PTDF coupling means NO2 sets the price everywhere, yielding a near-uniform 77 €/MWh across Scandinavian and Finnish zones. Only DK1/DK2 separate at 93 €/MWh. FBMC produces internal congestion that prevents full price equalisation, Norwegian zones range from 62.49 to 77.00 €/MWh, Swedish and Finnish zones cluster near 69-70 €/MWh, and Denmark stays at 93 €/MWh.

iii) Transmission flows and congestion rents. Under FBMC, five AC lines bind: NO4→SE1 (700 MW), NO3→NO1 (500 MW), NO5→NO2 (600 MW), SE4→DK2 (1300 MW), and SE3→DK1 (680 MW), plus two HVDC links: NO2→DK1 (1632 MW) and DK1→DK2 (600 MW). ATC binds seven AC lines, most notably NO5→NO1 at 3900 MW, a flow FBMC prohibits due to parallel-path effects.

This is clearest on NO5→NO1 and NO2→NO1: ATC routes 3900 MW from NO5 to NO1 with NO2→NO1 reversed at −1320 MW, while FBMC carries only 1749 MW on NO5→NO1 and 1595 MW forward on NO2→NO1. Congestion rent on each line is:

$$CR_l = f_l \cdot (\lambda_{to} - \lambda_{from}) \quad (24)$$

Table A.2 presents the most significant congestion rents and Figure A.4 shows the shadow prices of the binding AC constraints under FBMC. FBMC total congestion rent (107 258 €) is nearly double the ATC figure (57 792 €). Under ATC, rents arise only on Danish interconnectors since all other zones price at 77 €/MWh. FBMC generates additional rents on Norwegian internal lines due to PTDF-driven price differentiation; NO5→NO2 carries the highest shadow price at 30.96 €/MW, while SE4→DK2 and SE3→DK1 each show 23.00 €/MW, reflecting the mainland-to-Denmark price gap.

c) In theory, FBMC enables more efficient cross-zonal trade than ATC by replacing conservative NTC margins with PTDF-based constraints that expose additional feasible capacity. Three reasons explain why ATC produces the lower cost here.

Modelling assumptions. FBMC's advantage relies on TSOs setting ATC limits conservatively to guard against loop flows. PTDF constraints then reveal additional trading capacity. Both methods use identical line ratings here, so ATC is never penalised by conservative margins and that mechanism is absent.

Network representation. Single-node zones mean PTDFs are exact and no approximation error separates the two methods. The 12-node topology also lacks the dense parallel-path structure where loop-flow effects, and thus FBMC's advantage are most pronounced.

Constraint structure. PTDF coupling restricts dispatch combinations that would overload parallel paths. ATC's independent line flows face no such coupling, so its feasible region is strictly larger. Concretely, ATC routes 3900 MW on NO5→NO1 while reversing NO2→NO1 at −1320 MW. FBMC identifies this as a NO5→NO2 violation, forcing 2315 MW to shift from NO5 (63 €/MWh) to NO2 (77 €/MWh), directly producing the 27 060 € gap.

The results do not contradict theory. Identical capacity limits, single-node zones, and absent NTC conservatism collectively remove the conditions under which FBMC's advantages materialise.

3.3 Task 3-3: Dry-Year Scenario Comparison

a) Low hydro inflow sharply reduces Norwegian hydropower capacity while demand is unchanged. Total system cost rises to 5 062 341 € (FBMC) and 5 056 200 € (ATC), roughly 54% above the wet-year FBMC figure, reflecting greater reliance on expensive thermal and Danish generation.

Generation. Table A.3 and Figure A.5 report generation and prices for both methods. Nine of twelve zones produce at full capacity under FBMC: NO4, NO5, NO2, NO1, SE1, SE2, SE3, SE4, and FI. The wet-year surplus has largely vanished, NO5 capacity falls from 6500 to 3400 MW against unchanged 2600 MW demand, pushing nearly every zone to its limit. Only NO3 retains marginal spare capacity (3229 of 3300 MW), constrained by the binding NO3→NO1 line (500 MW), while DK1 and DK2 have headroom as power now flows into Denmark via the reversed NO2→DK1 HVDC link. Under ATC the dispatch is nearly identical, except DK1 produces 1900 MW against DK2's full 2100 MW, a routing difference that disappears without PTDF coupling.

Nodal prices. ATC yields a uniform 114 €/MWh across all twelve zones, with no binding constraint separating prices. FBMC produces several distinct clusters:

- **NO3 at 78 €/MWh:** the only zone below full capacity. Marginal demand is served locally.
- **NO4 at 85.27 €/MWh:** at full capacity with congested export lines, priced above its 52 €/MWh marginal cost.
- **SE1 (92.44), SE2 (94.35), SE3/SE4/FI (97.83 €/MWh):** all at full capacity, priced by the cost of importing the marginal MW through the congested AC network.
- **NO5, NO2, NO1, DK1 at 114 €/MWh:** expensive Danish generation sets the marginal price across this connected cluster.

- **DK2 at 108 €/MWh:** isolated from the 114 €/MWh cluster by the binding SE4→DK2 line.

Prices are substantially higher than the wet year (FBMC: 62- 93 €/MWh), with the floor shifting upward by 15-20 €/MWh as cheap hydro supply contracts.

b) Figure A.6 compares the AC line flows under FBMC and ATC for the dry year.

Binding constraints. Under FBMC, four AC lines bind: NO3→NO1 (500 MW), SE1→FI (1500 MW), SE4→DK2 (1300 MW), and SE3→DK1 (680 MW), fewer than the wet year's five, as reduced Norwegian surplus releases the NO4→SE1 and NO5→NO2 corridors. Under ATC, seven lines bind, including NO1→SE3 at full reverse capacity (−2095 MW) and NO5→NO2 at −500 MW, reflecting ATC's freedom to route large reverse flows without PTDF checks.

Flow pattern changes. The most notable dry-year changes under FBMC are:

1. **NO2→DK1 HVDC reverses.** The wet-year export of 1632 MW flips to −651 MW, Denmark now exports to Norway. DK1→DK2 reverses similarly (+600 to −600 MW).
2. **NO1→SE3 reverses.** A modest 144 MW flow becomes −1149 MW, as SE3 exports heavily toward NO1 to compensate for reduced Norwegian hydro.
3. **NO5→NO2 reverses.** The wet-year +600 MW surplus flow becomes −101 MW as NO5 loses its excess capacity.
4. **SE2→SE3 remains dominant.** Flow eases from 7285 MW to 6629 MW but stays the largest single flow in the system.

Congestion rents. ATC congestion rent is 0 €, uniform pricing at 114 €/MWh leaves no price differential across any line. Under FBMC, total rent is 87 972 €, below the wet-year figure of 107 258 € despite higher absolute prices. Two effects explain this: the wet year's large spread between cheap Norwegian zones (62-63 €/MWh) and Denmark (93 €/MWh) has compressed, with Norwegian prices now at 78-114 €/MWh, and fewer AC lines are saturated, with several high-flow corridors carrying smaller volumes. The rent distribution shifts accordingly, SE2→SE3 and NO1→SE3 now dominate (23 070 and 18 586 €), while NO3→NO1 generates 18 000 € from its 36 €/MWh price gap despite only 500 MW of flow.

c) Figure A.7 compares FBMC zonal prices and net positions between the wet and dry year. Table A.5 aggregates results by country.

Nodal price changes. All zonal prices rise, but the magnitude differs. Norwegian zones see the largest increases: NO5 by 51 €/MWh (63 to 114 €/MWh), NO1 by 41 €/MWh, NO2 by

37 €/MWh, a demand-weighted average of +34 €/MWh (+49%). This is expected: Norway's mix is almost entirely hydropower, so reduced inflow directly removes its cheapest capacity. Swedish prices rise by 27 €/MWh (+39%) as nuclear and thermal capacity in SE3 absorbs some of Norway's reduced surplus at higher cost. Finland tracks the SE3/SE4/FI cluster at 97.83 €/MWh (+28 €/MWh), reflecting dearer imports through the SE1→FI and SE3→FI corridors. Danish prices rise by only 18.6 €/MWh (+20%), the smallest shift since Denmark's thermal-dominated mix is unaffected by hydro inflow. The increase reflects reduced Scandinavian import availability alone.

Norway reverses from net exporter to net importer. Norwegian generation falls from 24 676 MW to 19 729 MW against unchanged demand of 21 600 MW, swinging the net position from +3076 MW to −1871 MW, a shift of nearly 5000 MW. Sweden covers most of the deficit, raising net exports from +2236 MW to +4900 MW. Denmark's own deficit shrinks from −3612 MW to −1329 MW as the NO2→DK1 HVDC link reverses.

Role of interconnection. Interconnection is critical in both years, but for opposite reasons:

- **Wet year:** Norway's 3076 MW surplus cannot be absorbed domestically. Without interconnectors, cheap hydro would be curtailed while neighbours rely on expensive local generation. Cross-border capacity converts this surplus into exports and congestion rent revenue.
- **Dry year:** Norway's 1871 MW deficit requires imports for supply security. Without interconnectors, load shedding at 3200 €/MWh would be unavoidable. The reversed NO2→DK1 flow (−651 MW) and the SE3→NO1 import (−1149 MW) demonstrate interconnectors functioning as a security-of-supply lifeline.

Norway's 49% price increase, far exceeding any other country, underscores this asymmetry: weak interconnection would amplify seasonal price volatility and risk physical shortages in dry years.

3.4 Task 3-4: Phasing Out Baseload Production

a) Table A.6 summarises the objective values and Figure A.8 presents the generation dispatch and zonal prices. Total system cost rises by approximately 9.16 million € (+278%), from 3.29 to 12.45 million €, driven by two reinforcing factors.

Insufficient total generation capacity. The base case has zero reserve margin, total capacity (60 600 MW) exactly meets demand. Removing 8400 MW of nuclear from SE3 leaves 52 200 MW against unchanged demand of 60 600 MW. Even with every remaining generator at full

output, a system-wide shortfall of 8400 MW persists. Transmission constraints prevent full utilisation of remaining capacity, so the actual load shedding is 2862 MW, smaller than the raw capacity gap because some zones previously ran below their limits.

Load shedding at extreme cost. Shedding occurs at SE4 (2400 MW), FI (362 MW), and DK2 (100 MW), each at 3200 €/MWh, contributing $2862 \times 3200 \approx 9.16$ million €, virtually the entire cost increase. The generation component changes only modestly: Norwegian generators are dispatched more heavily (NO5: +739 MW, NO2: +1678 MW, DK1: +1420 MW, DK2: +1200 MW), partially offset by small reductions at NO4 (−231 MW) and NO3 (−33 MW).

Why shedding concentrates where it does. SE4 sheds 2400 MW because its 1600 MW local capacity cannot cover 4000 MW demand, and the SE3→SE4 corridor drops to zero once SE3 can barely serve its own load (4000 MW generation vs. 14 100 MW demand). Finland sheds 362 MW because its import corridors, SE1→FI (1500 MW) and the SE3→FI HVDC (1200 MW), cannot compensate for reduced Swedish output. Norwegian zones avoid shedding entirely, as sufficient local hydro and closer electrical proximity to the remaining surplus keep them balanced.

b) Figure A.9 shows the change in generation at each zone relative to the base case. Table A.7 summarises the dispatch comparison.

SE3 loses 8400 MW of nuclear (from 12 400 to 4000 MW), and the system compensates by increasing output at every available generator. Replacement power comes from four main sources:

1. **Southern Norway (NO2)** provides the largest increase (+1678 MW, to 9905 MW). With a 10 000 MW cap, NO2 has the most spare capacity and exports heavily through NO2→NO1→SE3. The NO1→SE3 flow rises from 144 to 2145 MW, hitting its limit.
2. **Denmark (DK1, DK2)** increases by 1420 and 1200 MW. Previously a net importer, Denmark now ramps expensive thermal generation (93 €/MWh) to export toward Sweden. SE3→DK1 reverses from +680 to −740 MW.
3. **Western Norway (NO5)** increases by 739 MW, drawing on unused hydro capacity. NO5→NO1 rises from 1749 to 2488 MW to push power toward SE3.
4. **SE4** reaches full capacity (+764 MW, to 1600 MW), but cannot cover its own 4000 MW demand, with SE3→SE4 dropping to zero, SE4 sheds 2400 MW.

NO4 and NO3 decrease slightly (−231 and −33 MW): PTDF constraints mean increasing output there would overload parallel paths, so the optimiser shifts generation to zones with better electrical connectivity to the SE3 deficit.

Despite the system-wide response, 2862 MW goes unserved — SE4 (2400 MW), FI (362 MW), and DK2 (100 MW), all electrically downstream of the SE3 bottleneck and unable to receive sufficient imports through the congested network.

c) Figure A.10 compares zonal prices before and after the nuclear phaseout. The phaseout splits the system into two sharply separated price zones.

Shedding-price zones (3200 €/MWh). SE1, SE3, SE4, FI, and DK2 rise to the load-shedding penalty. SE2 reaches 3104 €/MWh, slightly below due to PTDF coupling with both the shedding cluster and Norwegian zones. Even SE1 and SE2, which shed no load themselves, price at near-shedding levels because an additional MW of demand there would increase flows toward SE4 or FI, triggering more shedding at 3200 €/MWh.

Norwegian and DK1 zones (27-93 €/MWh). Norwegian prices fall or hold: NO4 drops from 62.49 to 27 €/MWh, NO3 from 63.74 to 38 €/MWh, while NO5 (63), NO2 (77), and NO1 (72.72 €/MWh) are unchanged. DK1 stays at 93 €/MWh. These zones are separated from the shedding cluster by saturated constraints, NO1→SE3 and NO4→SE1 both bind, so marginal Norwegian demand is served by cheap local generation rather than triggering further Swedish shedding. The NO4 and NO3 price drops reflect the changed flow pattern: bottlenecks on their export corridors that previously lifted prices above marginal cost are released after the phaseout, returning both zones to their own marginal costs.

Who pays the most. Consumers in SE3, SE4, FI, and SE1 face 3200 €/MWh, a 45-fold increase from the base-case 70 €/MWh. SE4 is hardest hit: 60% of its demand (2400 of 4000 MW) goes unserved, with remaining load clearing at the shedding price. Finland faces 362 MW unserved at the same spike.

Least affected. Norwegian zones are shielded by sufficient local hydro and the saturated NO1→SE3 corridor (2145 MW): Norway exports everything it can, but the congested interconnector prevents the Swedish supply crisis from transmitting into Norwegian prices. DK1 is equally unaffected at 93 €/MWh throughout.

d) The results show that removing 8400 MW of baseload without preparation causes shedding, extreme price spikes, and a fractured price landscape. Several measures would need to precede such a policy.

1. Build replacement capacity near the deficit zone. Total system capacity falls below demand, making shedding inevitable regardless of network configuration. New capacity, wind, solar, or gas, must be located in or near SE3 specifically: Norwegian and Danish surplus cannot fully substitute due to transmission bottlenecks, so location matters as much as total installed capacity.

2. Expand key transmission corridors. Even with sufficient system-wide generation, bind-

ing constraints prevent power reaching deficit zones. NO1→SE3 saturates at 2145 MW and SE3→SE4 drops to zero. Reinforcing these corridors and the SE3→FI HVDC would allow surplus to reach SE4 and Finland. The base-case shadow prices already flag these as the highest-value bottlenecks.

3. Phase the retirement gradually. Removing all 8400 MW simultaneously creates an immediate shock. A multi-year phaseout generates incremental price signals that incentivise replacement investment and demand-side adjustment before each retirement step.

4. Develop demand-side flexibility and storage. All demand in the model is inelastic, so shortfalls translate directly into involuntary shedding at 3200 €/MWh. Interruptible contracts and real-time pricing for industrial consumers in SE4 and Finland would achieve voluntary reductions at far lower cost. Batteries and pumped hydro could buffer short-term imbalances.

5. Coordinate across Nordic TSOs. Finnish consumers face shedding despite no change in their own generation, purely through import dependence on SE3. Coordinated capacity planning, reserve sharing, and agreed flow protocols between Nordic TSOs are essential to ensure neighbouring countries are prepared for the changed dispatch and flow patterns.

Without these preconditions, new generation, transmission reinforcement, gradual implementation, and enhanced flexibility, the burden falls disproportionately on Sweden, Finland, and parts of Denmark.

3.5 Task 3-5: Environmental Constraint

a) Emission intensity values were obtained from Electricity Maps at 12:00 CET on 27 March 2026. A single intensity is applied to all Swedish zones. Table A.8 lists the values alongside the wet-year FBMC dispatch.

Finland is by far the largest emitter: 931 tCO₂/h (44% of system total) despite producing only 16% of generation, reflecting its 95 gCO₂/kWh mix of gas, peat, and biomass CHP. Denmark is second: DK1 at 155 gCO₂/kWh (the highest intensity in the system) contributes 231 tCO₂/h, DK2 a further 36 tCO₂/h. SE3 ranks third in absolute terms (186 tCO₂/h) solely due to its large volume (12 400 MW); at 15 gCO₂/kWh it has the lowest intensity, reflecting a nuclear and hydro mix. Norway contributes 541 tCO₂/h at an average 22 gCO₂/kWh, almost entirely lifecycle emissions from hydropower infrastructure. Absolute emissions depend on both intensity and volume: Finland emits more than Denmark despite lower intensity because it generates far more electricity (9800 vs. 1788 MW).

The ranking is entirely consistent with the underlying technology mix. The key insight is that absolute emissions depend on both intensity and volume: Finland emits more than Denmark

despite lower intensity because it generates far more electricity (9800 vs. 1788 MW). Similarly, SE3 has high absolute emissions despite the lowest intensity, simply because it is the largest single generating zone in the system.

b) The EU ETS has passed through four distinct phases.

Phase 1 (2005-2007): Over-allocated allowances caused prices to briefly reach ~ 30 €/tCO₂ before collapsing to near zero when the surplus became apparent and banking was prohibited.

Phase 2 (2008-2012): The financial crisis suppressed industrial emissions below the cap. Prices averaged ~ 15 €/tCO₂ and a large surplus accumulated.

Phase 3 (2013-2020): Prices remained depressed at 5-8 €/tCO₂ until the Market Stability Reserve (MSR) began absorbing surplus in 2019, driving prices from ~ 8 € in 2017 to ~ 35 €/tCO₂ by end-2020.

Phase 4 (2021-present): Prices surged above 95 €/tCO₂ in early 2023, driven by the tightened cap, continued MSR absorption, and the energy crisis following Russia's invasion of Ukraine. Prices have since moderated to 65-75 €/tCO₂; the 2021-2025 average is approximately 70-75 €/tCO₂.

Forecast to 2050. The trajectory points to sustained increase, driven by a linear reduction factor rising to 4.3% per year after 2024, phase-out of free allowances by 2034, and the Carbon Border Adjustment Mechanism (CBAM):

- **2026-2027:** 65-85 €/tCO₂, with limited upside pending the 2026 ETS directive review.
- **2030:** 100-150 €/tCO₂. BloombergNEF forecasts ~ 150 € for ETS2, with ETS1 at 100-120 €/tCO₂.
- **2040-2050:** 150-300 €/tCO₂ or higher, as the cap approaches zero by 2039 and remaining emissions require full offsetting.

The overall trend moves from near-zero in the pilot phase to potentially several hundred euros per tonne by mid-century, reflecting the progressively tighter carbon budget required to meet the EU's net-zero commitment.

c) The carbon cost is added to each zone's marginal cost:

$$c_n^{\text{new}} = c_n + e_n \cdot p_{\text{CO}_2} \quad (25)$$

where e_n [tCO₂/MWh] is the emission intensity and $p_{\text{CO}_2} = 65$ €/tCO₂. The resulting adders are shown in Table A.9.

Objective value. At base-case emissions of 2104 tCO₂/h, the gross additional cost is roughly 137 000 €/h. The actual increase will be somewhat lower as the optimiser re-dispatches toward cleaner units.

Generation share. Cost adders are highly asymmetric: Norwegian and Swedish zones see increases of only 0.97-1.69 €/MWh, while DK1 gains +10.08 €/MWh (to 103.08 €/MWh), DK2 +7.80 €/MWh, and Finland +6.18 €/MWh (to 65.18 €/MWh, overtaking NO5 at 64.24 €/MWh). The merit order is largely preserved, but two shifts are expected: Finland becomes marginally less competitive relative to Norwegian hydro, and Danish generation becomes the most expensive in the system, strengthening Denmark's incentive to import rather than generate locally.

Nodes expected to behave differently. DK1, DK2, and FI are most affected. Danish generation, already the most expensive in the system, becomes even less competitive, likely reducing local output and increasing imports via the HVDC links and the SE4→DK2 corridor. Finnish generation may decrease marginally if the network permits substitution with cheaper Norwegian hydro. Norwegian and Swedish zones are expected to behave near-identically to the base case, with cost increases below 2 €/MWh.

Congestion. Corridors to Denmark (SE4→DK2, SE3→DK1, NO2→DK1 HVDC) may see reduced congestion if higher Danish costs narrow the price differential driving congestion rents. Conversely, SE1→FI could tighten if Finnish output is displaced by cheaper Scandinavian imports. Internal Norwegian and Swedish corridors are unlikely to change materially.

d) With EU ETS prices forecast to reach 100-150 €/tCO₂ by 2030 and 150-300 € by 2040-2050, four trends follow for the Nordic system.

Prices. At 150 €/tCO₂, DK1's adder reaches 23.25 €/MWh (effective cost ~116 €/MWh), Finland gains 14.25 €/MWh, while Norwegian and Swedish zones see only 2-4 €/MWh. The price gap between clean and carbon-intensive zones widens, further advantaging hydro- and nuclear-dominated regions.

Generation share. Danish thermal faces growing displacement by imports and wind, potentially becoming entirely uneconomic at high carbon prices. Finnish CHP shifts away from peat and gas toward nuclear and wind. Norwegian and Swedish hydro/nuclear become more competitive, with stronger export incentives.

Total system cost. Cost increases roughly linearly with the carbon price if the mix is fixed. In practice, new wind and nuclear investment will reduce emission intensity over time, partially offsetting the rise. The net effect is moderate cost growth alongside significant emission reductions.

Congestion and long-term equilibrium. Wider price differentials intensify congestion on

corridors to Denmark and Finland (SE4→DK2, SE3→DK1, NO2→DK1, SE1→FI, SE3→FI HVDC), generating stronger signals for transmission expansion. At 200-300 €/tCO₂, near-complete decarbonisation becomes inevitable. Once all generation approaches zero emissions, carbon adders converge to zero and prices revert to fuel costs and scarcity.

e) A carbon price of 65 €/tCO₂ was implemented by adding $e_n \cdot p_{\text{CO}_2}$ to each zone's marginal cost in the FBMC model, using intensities from Table A.8. Results are compared against the base case in Table A.10 and Figure A.12.

1. Generation dispatch: the Danish swap. Dispatch is identical across all ten Scandinavian and Finnish zones. DK1 falls from 1488 to 288 MW while DK2 rises from 300 to 1500 MW. Both zones shared a marginal cost of 93 €/MWh in the base case; after the CO₂ adder, DK1 (155 gCO₂/kWh) reaches 103.08 €/MWh against DK2's 100.80 €/MWh (120 gCO₂/kWh), so the optimiser substitutes DK1 with DK2, with the DK1↔DK2 HVDC reversing from +600 to −600 MW.

2. AC flows unchanged. All fifteen AC flows are identical to the base case. Since the AC network reaches Denmark only through SE4→DK2 and SE3→DK1, and the total Danish net position seen by the AC network is unchanged, the five binding constraints from Task 3-2 remain the same.

3. Zonal prices. Prices increase by approximately their local adder: Norwegian zones by 0.07-1.43 €/MWh, Swedish and Finnish by ~1 €/MWh, Denmark by 7.80-10.08 €/MWh, confirming the marginal generator at each node is unchanged.

4. System cost and emissions. Total cost rises by 134 037 € (+4.1%), from 3.29 to 3.43 million €. Emissions fall by only 42 tCO₂/h (−2.0%), as 65 €/tCO₂ is insufficient to displace Finnish or other thermal generation, the merit order change is confined to the Danish swap.

5. Unanticipated result. The DK1↔DK2 swap was not predicted in part (c). Both Danish zones were previously interchangeable at 93 €/MWh. The carbon price breaks this symmetry by penalising DK1 more heavily, creating an intra-country differential absent from the base case. This shows carbon pricing can split zones that were previously identical, and that HVDC links are essential for enabling the resulting re-optimisation.

4 Problem 4: Wind Integration and Price Spikes

4.1 Task 4-1: Wind Integration in the Base (Wet-Year) Case

Wind is modelled as a reduction in net demand at each zone: $D_n^{\text{net}} = D_n - W_n$, where W_n is installed wind capacity. At zero marginal cost, wind is always dispatched first, so the two modelling approaches (separate generator or demand reduction) yield identical results. This preserves the existing model structure without additional decision variables. Total wind capacity is 19 100 MW; DK1 and DK2 wind exactly matches their demand, leaving zero net demand in both zones.

1. Zonal generation and prices. Wind displaces 19 100 MW of conventional generation (total conventional output falls from 60 600 to 41 500 MW). The largest reductions are at the most expensive units: NO2 (−5310 MW), SE3 (−5394 MW), and FI (−3 300 MW). Danish conventional generation is eliminated entirely. Zonal prices fall across the board; the demand-weighted average drops from 71.92 to 58.24 €/MWh (−19%). The largest decreases are at DK2 (−38 €/MWh), NO4 (−35.49 €/MWh, falling to its own marginal cost of 27 €/MWh), and SE1 (−27.45 €/MWh).

2. Merit-order effect. Zero-cost wind shifts the supply curve rightward, displacing the most expensive conventional generators. In the base case, SE4 (70 €/MWh) was marginal in the Nordic/Swedish cluster; with wind, SE3 (55 €/MWh) becomes marginal in several zones. NO4 and NO3 fall to their own marginal costs (27 and 38 €/MWh) as reduced net demand removes the need for congestion-elevated export pricing.

3. AC and HVDC flow changes.

- NO2→NO1 drops from 1595 to 297 MW, reflecting reduced NO2 conventional output.
- NO1→SE3 reverses from +144 to −426 MW. Reduced Swedish net demand eliminates the need for Norwegian imports.
- SE3→SE4 halves from 4464 to 2100 MW as SE4's 2000 MW wind covers half its demand locally.
- NO2→DK1 HVDC reverses from +1632 to −1280 MW; with Danish demand fully covered by wind, the link exports Danish surplus back toward Norway.
- SE3→FI HVDC increases from 585 to 1200 MW (binding), as cheap Swedish power is pushed toward Finland.

4. Congestion. Three new binding constraints appear: NO4→NO3 (1200 MW), SE1→FI (1500 MW), and SE2→SE3 (7300 MW), along with the SE3→FI HVDC at 1200 MW. NO4→SE1 is relieved from 700 to 500 MW and SE4→DK2 from 1300 MW (binding) to 600 MW (non-binding). Wind relieves north-to-south export corridors but creates new bottlenecks on Finnish feed lines and internal Swedish lines.

5. Congestion rents. Total rent rises from 107 258 to 195 798 € (+83%). Despite lower average prices, wind widens zonal price differentials: prices now range from 27 to 77 €/MWh versus the tighter 69-77 €/MWh base-case cluster. SE2→SE3 alone generates 60 590 €, driven by an 8.3 €/MWh spread across 7 300 MW.

6. Total system cost. Cost falls by 1 223 498 € (−37.2%), from 3.29 to 2.07 million €, reflecting direct savings from replacing expensive conventional generation with zero-cost wind.

Why zero-cost wind reduces average prices but increases volatility. The merit-order effect guarantees lower prices when wind is available. During low-wind periods, however, the system reverts to conventional dispatch at higher prices, widening the spread between high- and low-wind hours. Wind-induced congestion shifts further widen zonal price differentials unpredictably, adding spatial volatility on top of temporal volatility. Lower average prices with higher variance is a well-documented consequence of large-scale wind integration.

4.2 Task 4-2: Price Spikes from Peak Demand

Two peak-demand scenarios scale demand at DK2 and SE4, building on the wind base case from Task 4-1. A system-wide +30 % stress test is added to illustrate the nonlinear price response near the capacity boundary.

- **Scenario A:** +10 % demand at DK2 and SE4
- **Scenario B:** +20 % demand at DK2 and SE4
- **Stress test:** +30 % demand system-wide

Scenario A and B results. In both scenarios no transmission constraint becomes newly binding and no zonal price changes relative to the wind base case (see Table A.12). SE3 is the marginal generator throughout, setting a price of 55 €/MWh across the Swedish, Finnish, and Danish clusters. The additional demand (620 MW in Scenario A; 1240 MW in Scenario B) is absorbed entirely by SE3, which retains approximately 5400 MW of spare capacity in the wind base case (12 400 MW cap vs. 7006 MW dispatch). The SE3→SE4 and SE3→DK2 corridors remain uncongested, so the marginal cost of serving the extra load is simply SE3's unchanged cost of

55 €/MWh. Total system cost rises linearly at $55 \times \Delta D$ €/MW with no price spike; this is the flat region of the residual supply curve.

1. Why Scenarios A and B show no price change. The system sits on a flat portion of the residual supply curve: SE3 has ample spare capacity and the corridors connecting it to SE4 and DK2 are uncongested. No generator reaches its capacity limit, no transmission line saturates, and the marginal resource therefore does not change. Cost rises linearly ($55 \times \Delta D$ €/MW) with no price spike.

2. Binding constraints as demand rises. Two critical thresholds emerge:

- **At +80 % DK2 and SE4:** SE3 output reaches 10 106 MW and SE3→SE4 approaches capacity. SE4 rises from 55 to 70 €/MWh and DK2 to 77 €/MWh, the first price movements, marking the transition to the steep portion of the supply curve.
- **At +30 % system-wide:** SE3 hits its 12 400 MW cap and SE1→FI saturates, forcing 150 MW of Finnish shedding. Finland spikes to 3200 €/MWh while most other zones rise to ~ 73 €/MWh.

3. Marginal technology. SE3 (55 €/MWh) sets the price for the Swedish/Finnish/DK2 cluster while it has spare capacity. Once fully dispatched, SE4 (70 €/MWh) or NO1 (67 €/MWh) become marginal, producing price jumps of 12–17 €/MWh. When all generators are exhausted, load shedding at 3200 €/MWh becomes the marginal resource, a $58\times$ price increase.

4. Why small demand increases cause disproportionately large price spikes. The residual supply curve has three distinct regimes:

- **Flat:** Spare SE3 capacity absorbs the first ~ 5400 MW of additional demand with zero price effect. Scenarios A and B remain in this region.
- **Steep:** Once SE3 is exhausted, the marginal unit jumps to SE4; a 27 % price increase is triggered by a single marginal MW.
- **Vertical (scarcity):** Total capacity exhausted; price rises discontinuously to the Value of Lost Load (3200 €/MWh).

Binding transmission constraints amplify this nonlinearity. Finland reaches shedding prices in the +30 % scenario not from a system-wide capacity shortage, but because SE1→FI (1500 MW) and SE3→FI HVDC (1200 MW) are both saturated. The combination of a steep residual supply curve and congested interconnectors produces the extreme price spikes characteristic of scarcity pricing.

4.3 Task 4-3: Price Spikes from Supply Scarcity

A supply scarcity scenario reduces wind to 25 % of installed capacity, combined with a 30 % Norwegian hydro reduction to simulate compound scarcity. Table A.13 summarises the four scenarios analysed.

1. Zonal prices. Reducing wind progressively reverses the merit-order effect from Task 4-1:

- **Full wind → low wind (25 %):** Demand-weighted average rises from 58.24 to 63.78 €/MWh (+9.5 %). FI (59 €/MWh) becomes marginal in the eastern cluster; DK1/DK2 jump to 93 €/MWh as Danish thermal ramps up, and NO4 rises from 27 to 41 €/MWh.
- **Low wind → no wind:** Average rises to 71.92 €/MWh (+23.5 % vs. full wind), identical to the Task 3-2 base case — removing all wind returns the system to its pre-wind dispatch.
- **Compound scarcity (no wind + hydro –30 %):** All twelve zones spike to 3 200 €/MWh. Total capacity (59 700 MW) falls below demand (60 600 MW), forcing 900 MW of shedding at SE4; the shedding price propagates to every zone as no transmission constraint creates price separation.

2. Congestion pattern shifts. As wind decreases, congestion reverts toward the base-case structure: NO4→SE1 returns to binding (700 MW); SE3→FI HVDC falls from 1200 MW (binding, full wind) to 585 MW (non-binding); NO2→DK1 HVDC reverses from –1280 back to +1632 MW, restoring normal Norway-to-Denmark export. In the compound scenario, reduced Norwegian hydro shifts binding constraints to internal Swedish lines.

3. Congestion rents. In the compound scenario all zones share 3200 €/MWh, so every price differential is zero and total congestion rent collapses to approximately zero — the opposite of the full-wind case (195 798 €). The most extreme scarcity paradoxically produces zero rent. In intermediate scenarios, rents remain positive and follow the base-case pattern, with Danish corridors dominant at 93 €/MWh.

4. Largest price spikes. In the wind-reduction scenarios, Denmark sees the largest swings: DK1/DK2 rise from 55-77 €/MWh (full wind) to 93 €/MWh (no wind), a swing of 16-38 €/MWh, as zero-cost wind had been suppressing local thermal prices. In the compound scenario SE4 bears the physical impact (900 MW shed), but the shedding price propagates through the PTDF-coupled network to all twelve zones.

5. Do interconnectors mitigate or amplify scarcity? Both. SE3→SE4 delivers 4300 MW of imports, reducing SE4's potential 2400 MW shortfall to 900 MW. Finland avoids shedding entirely through SE1→FI and SE3→FI imports. Interconnectors therefore reduce total shedding volume. However, they simultaneously spread the 3200 €/MWh scarcity price to all twelve

zones, without interconnection, only SE4 would face shedding prices. Interconnectors mitigate physical scarcity while amplifying price exposure.

How low wind exposes marginal thermal pricing. With full wind, 19 100 MW of zero-cost generation suppresses prices to 27-77 €/MWh. Each lost MW of wind is replaced by the next merit-order unit: at 25 % wind, the marginal generator shifts from SE3 (55 €/MWh) to FI (59 €/MWh). At zero wind, SE4 (70 €/MWh) returns as marginal and prices revert to the pre-wind structure. The spread between full-wind conditions (NO4 at 27 €/MWh) and compound scarcity (3200 €/MWh system-wide) defines the volatility introduced by variable renewable integration, managing it through storage, demand response, and interconnection is one of the central challenges of the energy transition.

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A Appendix

A.1 Problem 3

A.1.1 3-2 b) ii)

Table A.1: Generation quantity and nodal prices (wet year).

Node	Name	Demand [MW]	GenCap [MW]	Cost [€/MWh]	Gen [MW]		Price [€/MWh]	
					FBMC	ATC	FBMC	ATC
1	NO4	2 700	4 600	27	4 600	4 600	62.49	77.00
2	NO3	4 000	3 900	38	3 900	3 900	63.74	77.00
3	NO5	2 600	6 500	63	4 949	6 500	63.00	77.00
4	NO2	5 600	10 000	77	8 227	5 912	77.00	77.00
5	NO1	6 700	3 000	67	3 000	3 000	72.72	77.00
6	SE1	1 500	4 500	35	4 500	4 500	70.00	77.00
7	SE2	2 500	6 600	35	6 600	6 600	69.41	77.00
8	SE3	14 100	12 400	55	12 400	12 400	70.00	77.00
9	SE4	4 000	1 600	70	836	1 600	70.00	77.00
10	FI	11 500	9 800	59	9 800	9 800	70.00	77.00
11	DK1	3 200	3 700	93	1 488	1 488	93.00	93.00
12	DK2	2 200	1 500	93	300	300	93.00	93.00

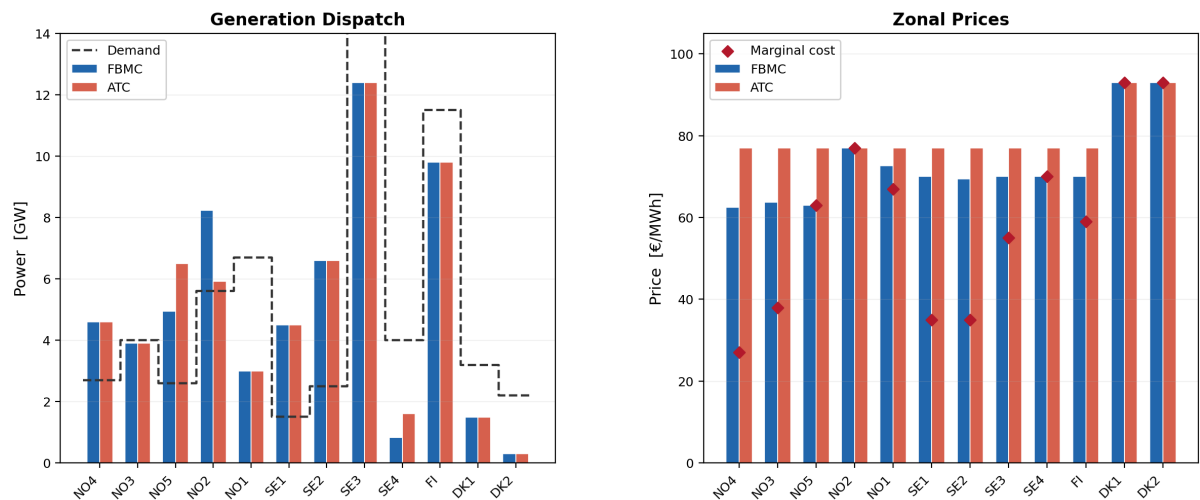


Figure A.1: Generation quantity (left) and nodal prices (right) under FBMC and ATC for the wet-year scenario. Red diamonds indicate the marginal generation cost at each zone.

A.1.2 3-2 b) iii)

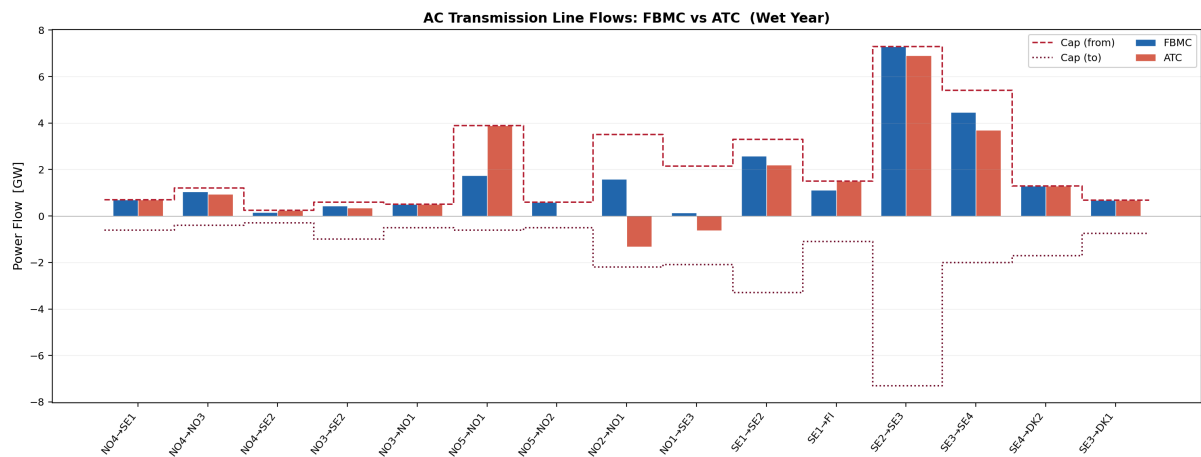


Figure A.2: AC transmission line flows under FBMC and ATC. Dashed red lines indicate the capacity limits in each direction.

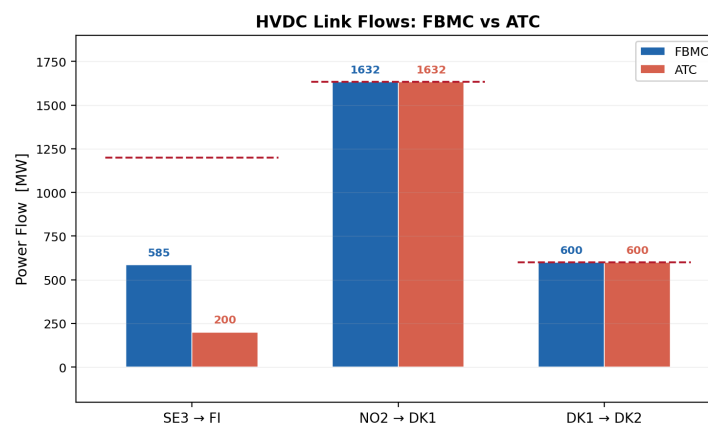


Figure A.3: HVDC link flows under FBMC and ATC.

Table A.2: Significant congestion rents (wet year). Only elements with $|CR| > 1000$ € are shown.

Line	Type	Flow [MW]	λ_{from}	λ_{to}	Rent [€]
<i>FBMC</i>					
SE4→DK2	AC	1 300	70.00	93.00	29 900
NO2→DK1	HVDC	1 632	77.00	93.00	26 112
NO5→NO1	AC	1 749	63.00	72.72	17 001
SE3→DK1	AC	680	70.00	93.00	15 640
NO5→NO2	AC	600	63.00	77.00	8 400
NO4→SE1	AC	700	62.49	70.00	5 257
NO3→NO1	AC	500	63.74	72.72	4 490
SE2→SE3	AC	7 285	69.41	70.00	4 298
Total FBMC congestion rent					107 258
<i>ATC</i>					
NO2→DK1	HVDC	1 632	77.00	93.00	26 112
SE4→DK2	AC	1 300	77.00	93.00	20 800
SE3→DK1	AC	680	77.00	93.00	10 880
Total ATC congestion rent					57 792

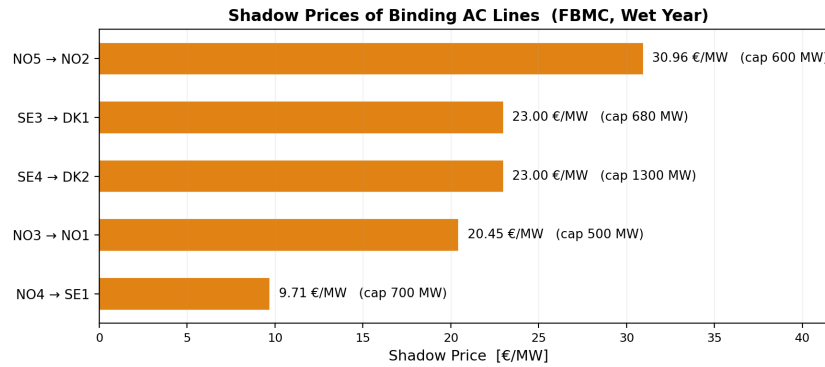


Figure A.4: Shadow prices of binding AC line constraints under FBMC (wet year). The shadow price represents the reduction in total system cost from a 1 MW increase in line capacity.

A.1.3 3-3 a)

Table A.3: Generation dispatch and zonal prices (dry year). GenCap shows the reduced dry-year capacity.

Node	Zone	Demand [MW]	GenCap [MW]	Cost [€/MWh]	Gen [MW]		Price [€/MWh]	
					FBMC	ATC	FBMC	ATC
1	NO4	2 700	3 900	52	3 900	3 900	85.27	114.00
2	NO3	4 000	3 300	78	3 229	3 300	78.00	114.00
3	NO5	2 600	3 400	96	3 400	3 400	114.00	114.00
4	NO2	5 600	7 500	98	7 500	7 500	114.00	114.00
5	NO1	6 700	1 700	99	1 700	1 700	114.00	114.00
6	SE1	1 500	4 200	61	4 200	4 200	92.44	114.00
7	SE2	2 500	8 000	60	8 000	8 000	94.35	114.00
8	SE3	14 100	12 900	87	12 900	12 900	97.83	114.00
9	SE4	4 000	1 900	96	1 900	1 900	97.83	114.00
10	FI	11 500	9 800	90	9 800	9 800	97.83	114.00
11	DK1	3 200	2 800	114	2 571	1 900	114.00	114.00
12	DK2	2 200	2 100	108	1 500	2 100	108.00	114.00

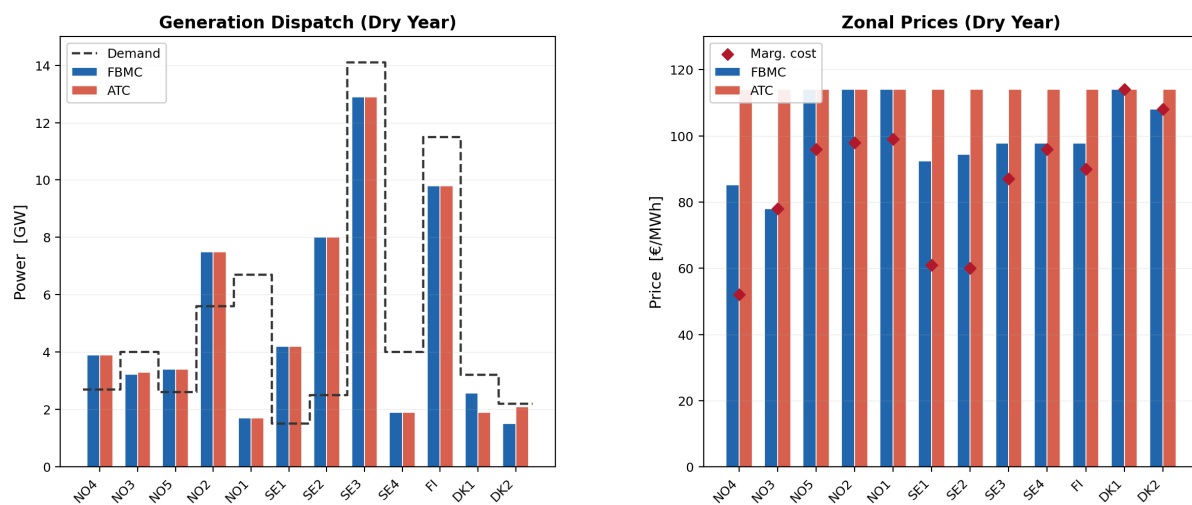


Figure A.5: Generation dispatch (left) and zonal prices (right) under FBMC and ATC for the dry-year scenario. Red diamonds indicate the marginal generation cost at each zone.

A.1.4 3-3 b)

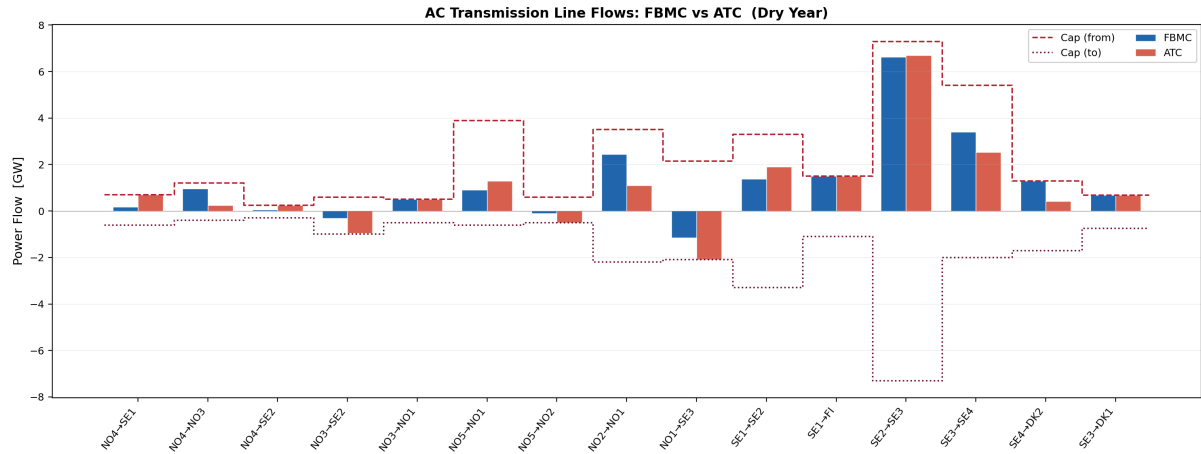


Figure A.6: AC transmission line flows under FBMC and ATC for the dry-year scenario. Dashed red lines indicate the capacity limits in each direction.

Table A.4: Significant congestion rents (dry year, FBMC). Only elements with $|CR| > 1000$ € are shown.

Line	Type	Flow [MW]	λ_{from}	λ_{to}	Rent [€]
SE2→SE3	AC	6 629	94.35	97.83	23 070
NO1→SE3	AC	−1 149	114.00	97.83	18 586
NO3→NO1	AC	500	78.00	114.00	18 000
SE4→DK2	AC	1 300	97.83	108.00	13 221
SE3→DK1	AC	680	97.83	114.00	10 996
SE1→FI	AC	1 500	92.44	97.83	8 085
DK1→DK2	HVDC	−600	114.00	108.00	3 600
SE1→SE2	AC	1 373	92.44	94.35	2 623
NO4→SE1	AC	173	85.27	92.44	1 241
Total FBMC congestion rent (dry year)					87 972

A.1.5 3-3 c)

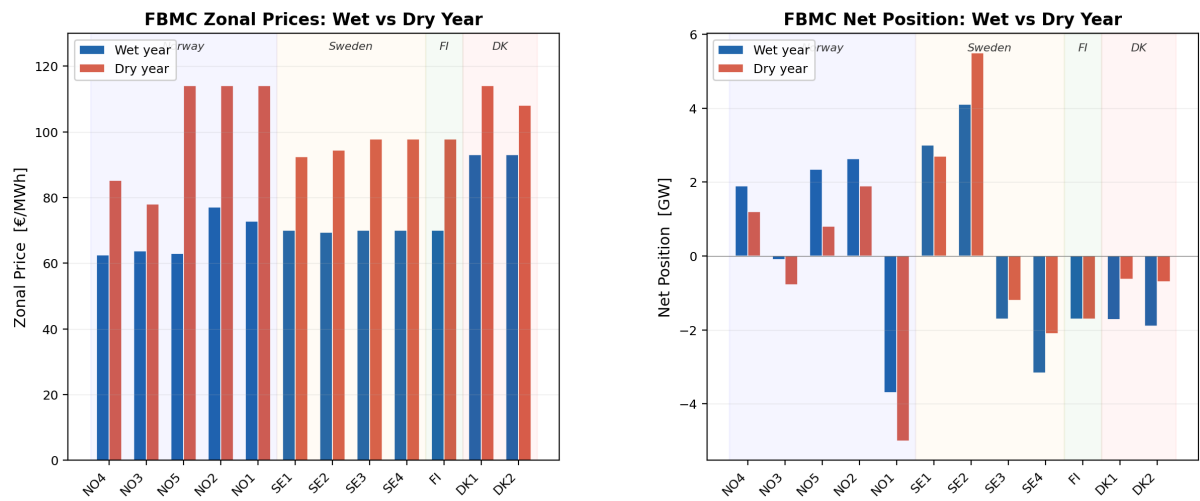


Figure A.7: FBMC zonal prices (left) and net positions (right) for the wet-year and dry-year scenarios. Positive net position indicates a net exporter.

Table A.5: Country-level aggregation of FBMC results (wet vs. dry year). Net position (NP) is total generation minus total demand. Price is demand-weighted average.

Country	Net Position [MW]		Avg. Price [€/MWh]		Price Change	
	Wet	Dry	Wet	Dry	Abs.	Rel.
Norway	+3 076	-1 871	69.7	103.7	+34.0	+49 %
Sweden	+2 236	+4 900	69.9	97.1	+27.1	+39 %
Finland	-1 700	-1 700	70.0	97.8	+27.8	+40 %
Denmark	-3 612	-1 329	93.0	111.6	+18.6	+20 %

A.1.6 3-4 a)

Table A.6: Total system cost: base case vs. nuclear phaseout (wet year, FBMC).

Scenario	Total Cost [€]		Change
Base case (wet)	3 292 168		—
Nuclear phaseout	12 454 629	+9 162 462	(+278 %)

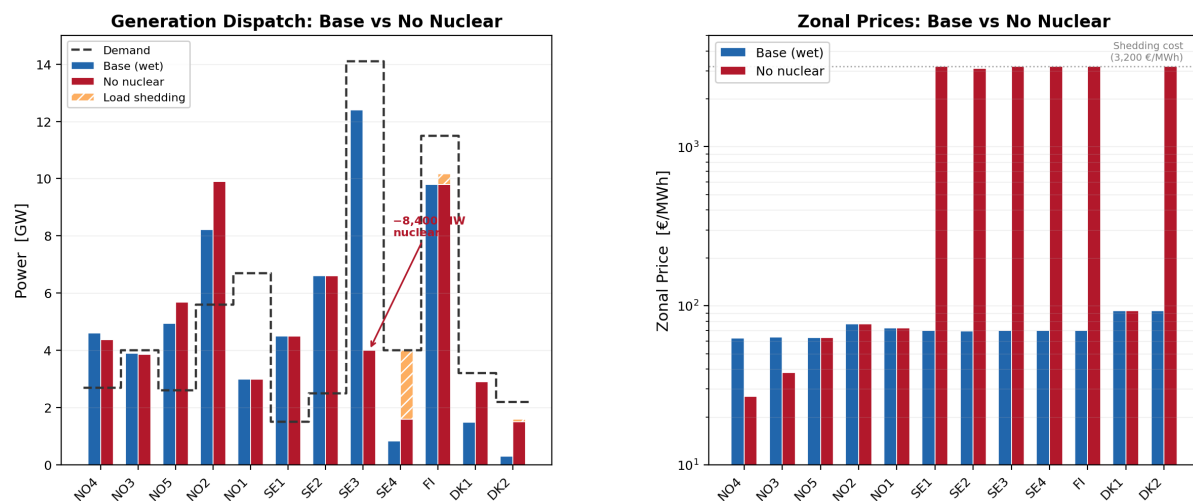


Figure A.8: Left: generation dispatch (with load shedding shown as hatched bars). Right: zonal prices on a logarithmic scale; the dashed line marks the shedding penalty of 3 200 €/MWh.

A.1.7 3-4 b)

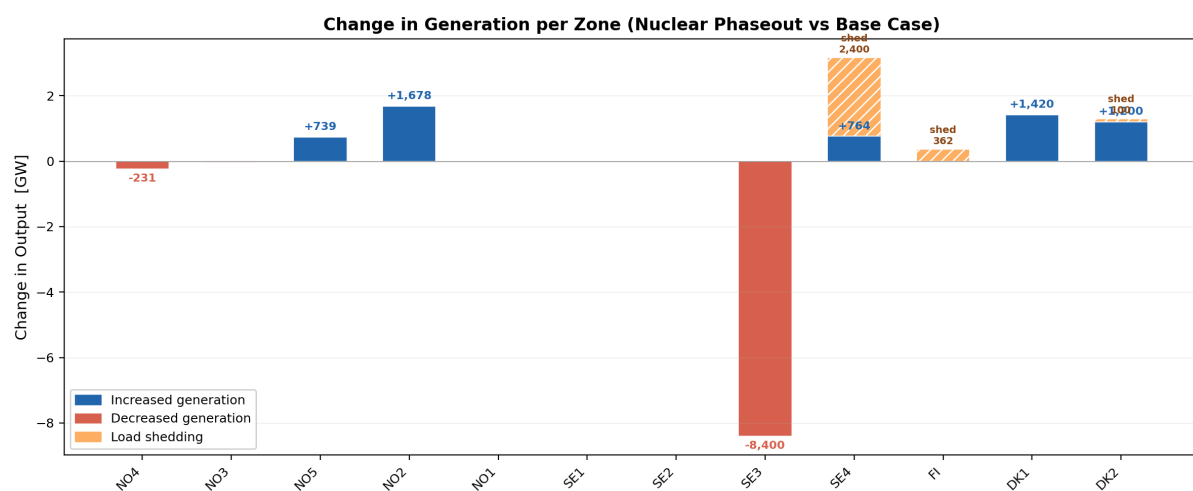


Figure A.9: Change in generation per zone after the nuclear phaseout. Blue bars indicate increased output, red bars indicate decreased output, and hatched orange bars indicate load shedding.

Table A.7: Generation dispatch comparison: base case vs. nuclear phaseout (FBMC).

Node	Zone	Base [MW]	No Nuke [MW]	Δ Gen [MW]	Shed [MW]	At cap?
1	NO4	4 600	4 369	−231	0	No
2	NO3	3 900	3 867	−33	0	No
3	NO5	4 949	5 688	+739	0	No
4	NO2	8 227	9 905	+1 678	0	No
5	NO1	3 000	3 000	0	0	Yes
6	SE1	4 500	4 500	0	0	Yes
7	SE2	6 600	6 600	0	0	Yes
8	SE3	12 400	4 000	−8 400	0	Yes
9	SE4	836	1 600	+764	2 400	Yes
10	FI	9 800	9 800	0	362	Yes
11	DK1	1 488	2 908	+1 420	0	No
12	DK2	300	1 500	+1 200	100	Yes
Total		60 600	57 738	−2 862	2 862	

A.1.8 3-4 c)

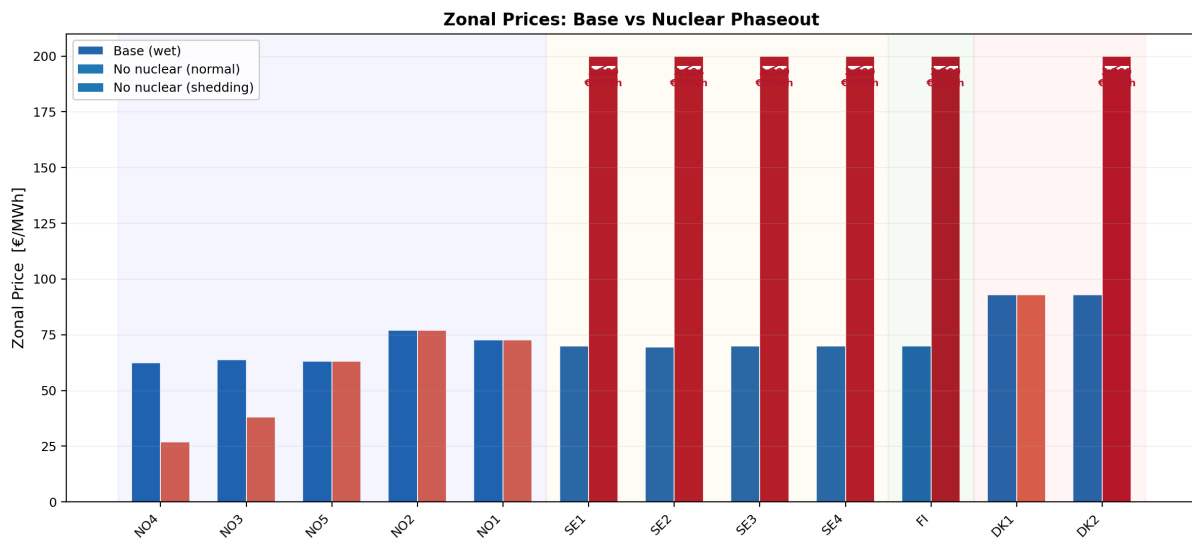


Figure A.10: Zonal prices: base case vs. nuclear phaseout. Bars capped at 200 €/MWh for readability; actual shedding prices (3200 €/MWh) are annotated in red.

A.1.9 3-5 a)

Table A.8: Carbon intensity of electricity generation by bidding zone, sourced from Electricity Maps (27 March 2026, 12:00 CET). Total emissions are computed for the wet-year FBMC base case dispatch.

Node	Zone	Gen [MW]	Intensity [gCO ₂ /kWh]	Emissions [tCO ₂ /h]	Share [%]
1	NO4	4 600	24	110.4	5.2
2	NO3	3 900	20	78.0	3.7
3	NO5	4 949	19	94.0	4.5
4	NO2	8 227	22	181.0	8.6
5	NO1	3 000	26	78.0	3.7
6	SE1	4 500	15	67.5	3.2
7	SE2	6 600	15	99.0	4.7
8	SE3	12 400	15	186.0	8.8
9	SE4	836	15	12.5	0.6
10	FI	9 800	95	931.0	44.2
11	DK1	1 488	155	230.6	11.0
12	DK2	300	120	36.0	1.7
Total		60 600	—	2 104	100

A.1.10 3-5 c)

Table A.9: Effect of a 65 €/tCO₂ carbon price on zonal marginal generation costs.

Node	Zone	Orig. Cost [€/MWh]	Intensity [gCO ₂ /kWh]	CO ₂ Adder [€/MWh]	New Cost [€/MWh]
1	NO4	27	24	1.56	28.56
2	NO3	38	20	1.30	39.30
3	NO5	63	19	1.24	64.24
4	NO2	77	22	1.43	78.43
5	NO1	67	26	1.69	68.69
6	SE1	35	15	0.98	35.98
7	SE2	35	15	0.98	35.98
8	SE3	55	15	0.98	55.98
9	SE4	70	15	0.98	70.98
10	FI	59	95	6.18	65.18
11	DK1	93	155	10.08	103.08
12	DK2	93	120	7.80	100.80

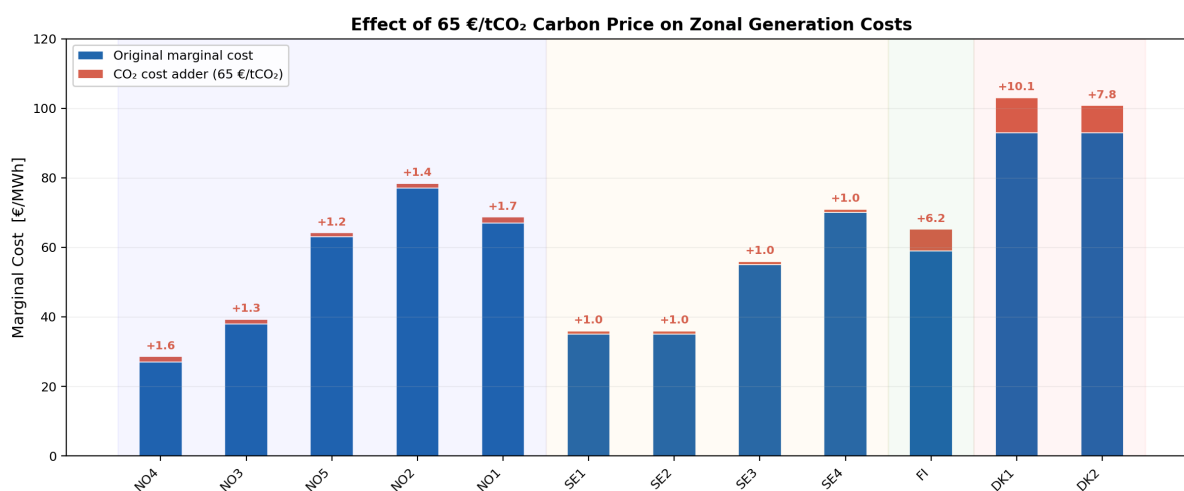


Figure A.11: Original marginal generation costs (blue) and the CO₂ cost adder (red) at each zone for a carbon price of 65 €/tCO₂.

A.1.11 3-5 e)

Table A.10: Comparison of FBMC results: base case vs. 65 €/tCO₂ carbon price (wet year).

Node	Zone	Gen [MW]		Price [€/MWh]		ΔPrice
		Base	CO ₂	Base	CO ₂	
1	NO4	4 600	4 600	62.49	62.37	−0.12
2	NO3	3 900	3 900	63.74	63.81	+0.07
3	NO5	4 949	4 949	63.00	64.24	+1.24
4	NO2	8 227	8 227	77.00	78.43	+1.43
5	NO1	3 000	3 000	72.72	74.09	+1.37
6	SE1	4 500	4 500	70.00	70.98	+0.98
7	SE2	6 600	6 600	69.41	70.30	+0.89
8	SE3	12 400	12 400	70.00	70.98	+0.98
9	SE4	836	836	70.00	70.98	+0.98
10	FI	9 800	9 800	70.00	70.98	+0.98
11	DK1	1 488	288	93.00	103.08	+10.08
12	DK2	300	1 500	93.00	100.80	+7.80
System cost		3 292 168		3 426 204		+4.1 %
Emissions [tCO₂/h]		2 104		2 062		−2.0 %

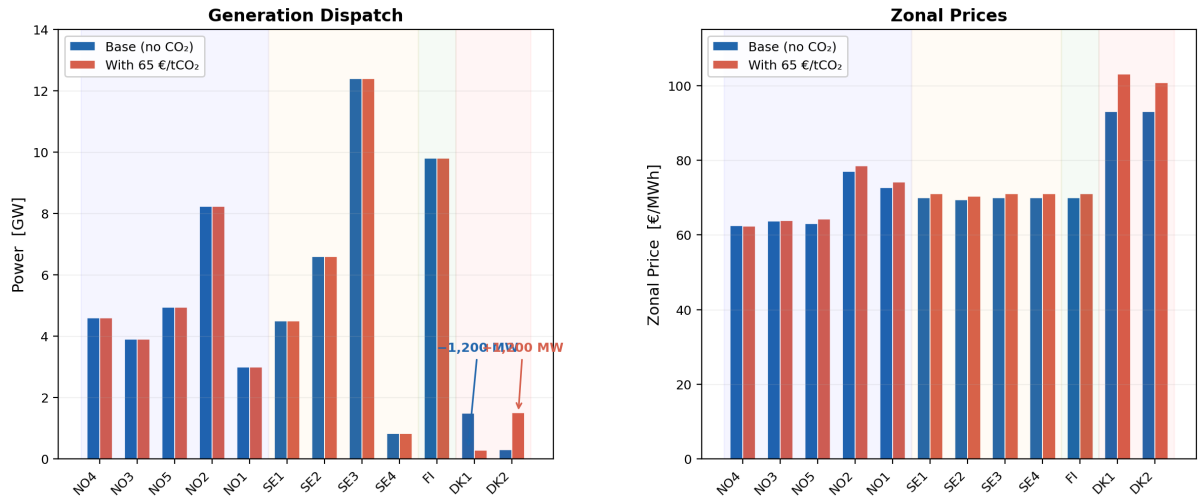


Figure A.12: Generation dispatch (left) and zonal prices (right) for the base case and the 65 €/tCO₂ scenario. The only dispatch change occurs in Denmark, where DK1 and DK2 swap roles.

A.2 Problem 4

A.2.1 4-1

Table A.11: Generation dispatch and zonal prices: base case vs. wind case (wet year, FBMC).

Node	Zone	Demand [MW]	Wind [MW]	Net D [MW]	Conv. Gen [MW]		Price [€/MWh]	
					Base	Wind	Base	Wind
1	NO4	2 700	500	2 200	4 600	4 029	62.49	27.00
2	NO3	4 000	800	3 200	3 900	2 571	63.74	38.00
3	NO5	2 600	500	2 100	4 949	3 877	63.00	63.00
4	NO2	5 600	1 100	4 500	8 227	2 917	77.00	77.00
5	NO1	6 700	1 300	5 400	3 000	3 000	72.72	72.72
6	SE1	1 500	400	1 100	4 500	4 500	70.00	42.55
7	SE2	2 500	600	1 900	6 600	6 600	69.41	46.70
8	SE3	14 100	4 200	9 900	12 400	7 006	70.00	55.00
9	SE4	4 000	2 000	2 000	836	500	70.00	55.00
10	FI	11 500	2 300	9 200	9 800	6 500	70.00	59.00
11	DK1	3 200	3 200	0	1 488	0	93.00	77.00
12	DK2	2 200	2 200	0	300	0	93.00	55.00
System cost [€]					3 292 168		2 068 670	
Avg. price [€/MWh]					71.92		58.24	

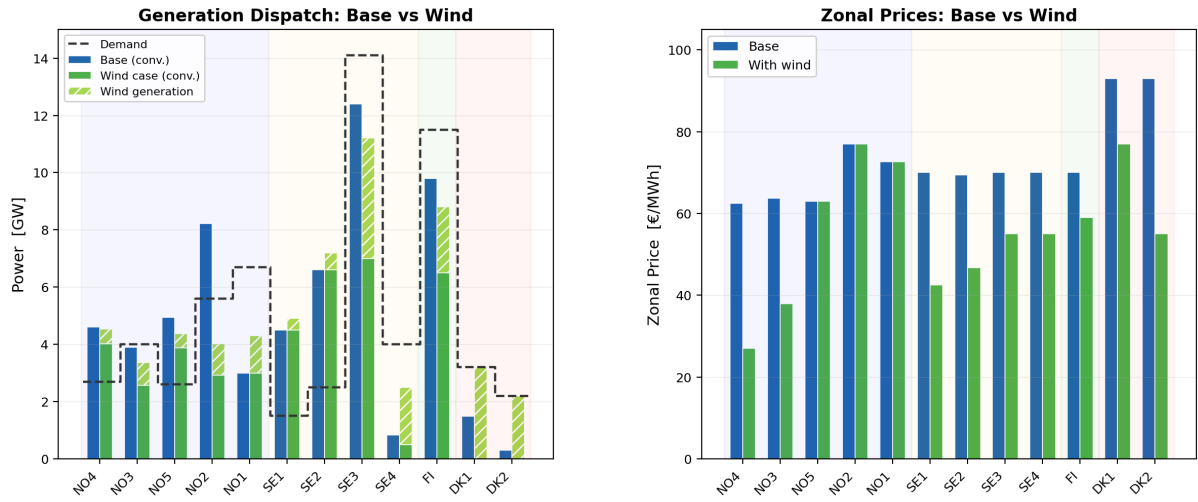


Figure A.13: Left: generation dispatch showing conventional output (solid) and wind (hatched). Right: zonal prices. Wind reduces both conventional output and prices across the system.

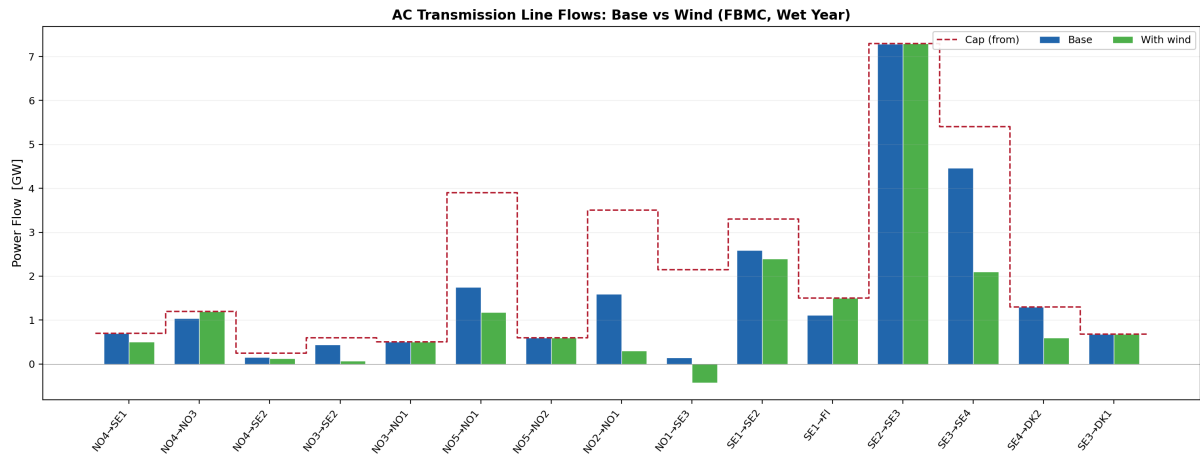


Figure A.14: AC line flows for the base case and wind case. Several corridors see reduced flows; two new lines become binding.

A.2.2 4-2

Table A.12: Peak demand scenario results compared to the wind base case (FBMC, wet year).

Scenario	Cost [€]	Shed [MW]	SE3 Gen [MW]	λ_{SE4}	λ_{FI}	λ_{DK2}
Base wind	2 068 670	0	7 006	55	59	55
+10% DK2, SE4	2 102 770	0	7 626	55	59	55
+20% DK2, SE4	2 136 870	0	8 246	55	59	55
+30% all zones	3 647 425	150	12 400	72.72	3 200	72.72

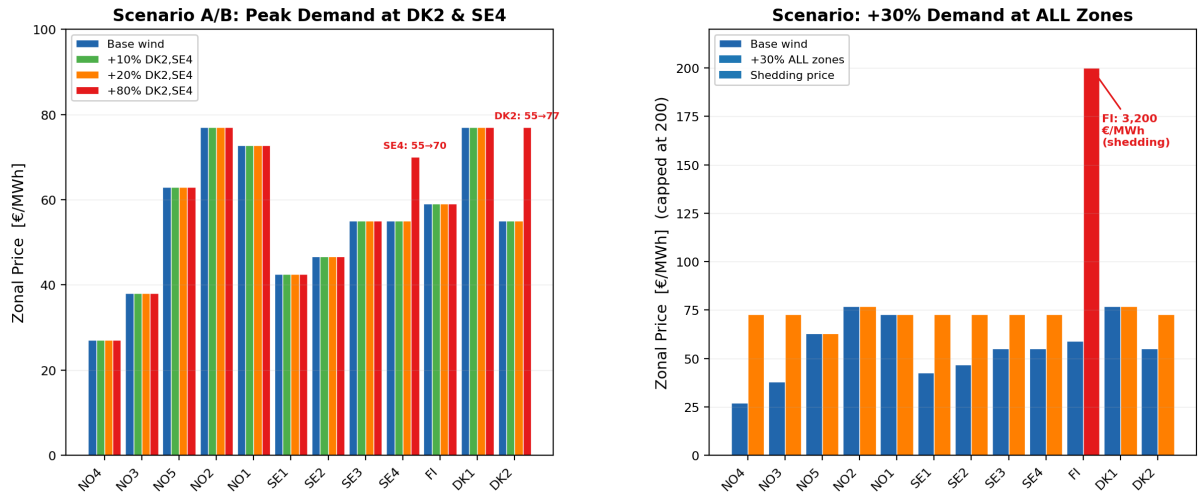


Figure A.15: Left: zonal prices under localised demand increases at DK2 and SE4 (Scenarios A and B remain identical to the base; +80 % shown to illustrate when SE4 begins to respond). Right: +30 % system-wide demand triggers shedding at Finland, with its price spiking to 3 200 €/MWh.

A.2.3 4-3

Table A.13: Supply scarcity scenario definitions and key results (FBMC, wet year).

Scenario	Description	Wind [MW]	Cost [€]	Shed [MW]	Avg. Price
Base wind	Full wind (100 %)	19 100	2 068 670	0	58.24
Low wind	Wind at 25 %	4 775	2 956 939	0	63.78
No wind	Wind at 0 %	0	3 292 168	0	71.92
Compound	No wind + hydro –30 %	0	6 281 330	900	3 200

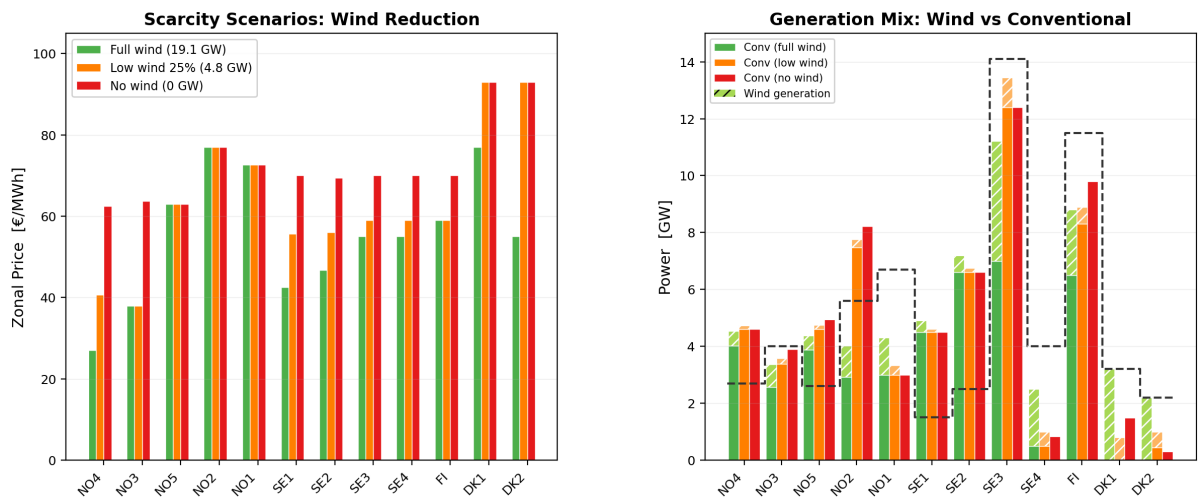


Figure A.16: Left: zonal prices across the three wind-availability scenarios (without hydro reduction). Right: conventional generation and wind output for each scenario.

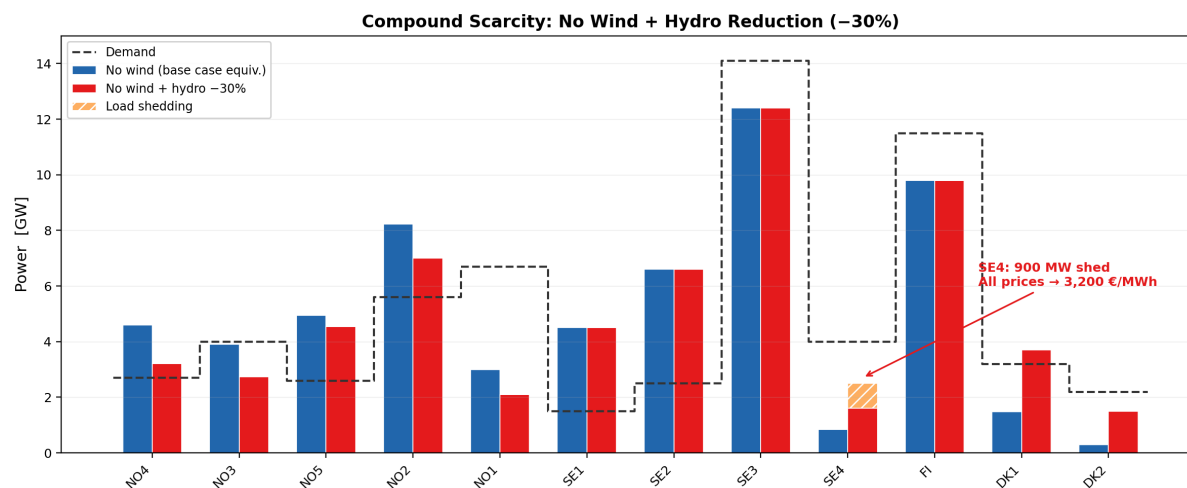


Figure A.17: Compound scarcity scenario (no wind + 30 % hydro reduction) compared to the no-wind base case. SE4 sheds 900 MW; all zonal prices spike to 3 200 €/MWh.